第二章　金属/电解质水溶液界面的基本性质与结构模型

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第二章 金属/电解质界面的基本性质与结构模型

3.1 电极/溶液界面研究意义

3.2 相间电势和电极电势

*静电学知识复习

实物相的电势 – 外电势、内电势(外电势+表面电势)和电化学势

电化学势与费米能级、相间电势、电极电势，氢标电势与费米能级

3.3 电化学界面的热力学

表面张力、表面超量、双电层电容

Gibbs 吸附等温式（电毛细现象）、电毛细方程

3.4 界面电参数和表面超量的实验测定

电毛细曲线、微分电容曲线、零电荷电位、表面超量

3.5 电极/溶液界面结构模型

电极/溶液界面基本图像 —紧密层和分散层

GCS 双电层理论 —界面电势分布

无离子特性吸附的紧密层模型 —溶剂层模型

离子特性吸附 —对界面电势分布和电极反应速率的影响

3.6 零电荷电位 **(PZC)**

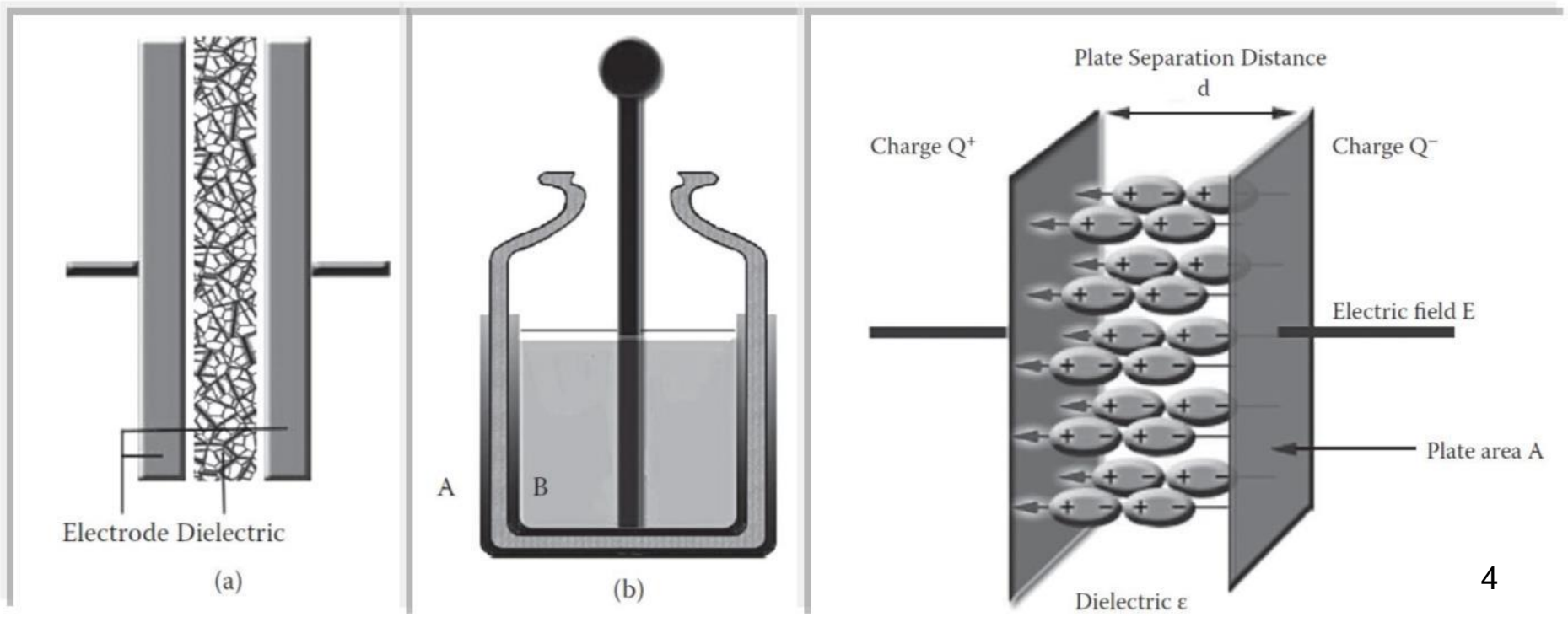
早期：Thales of Miletus 琥珀和布摩擦，引起表面储存电荷

18世纪中叶：静电学----起电盘（莱顿瓶---电容器）、静电感应电机

19世纪末、20世纪初：对电现象在分子原子级的理解

法拉第、汤姆逊、密立根

电子：带电荷量：4.80×10^{-10} esu 或 $1.602176462\times10^{-19}$ C



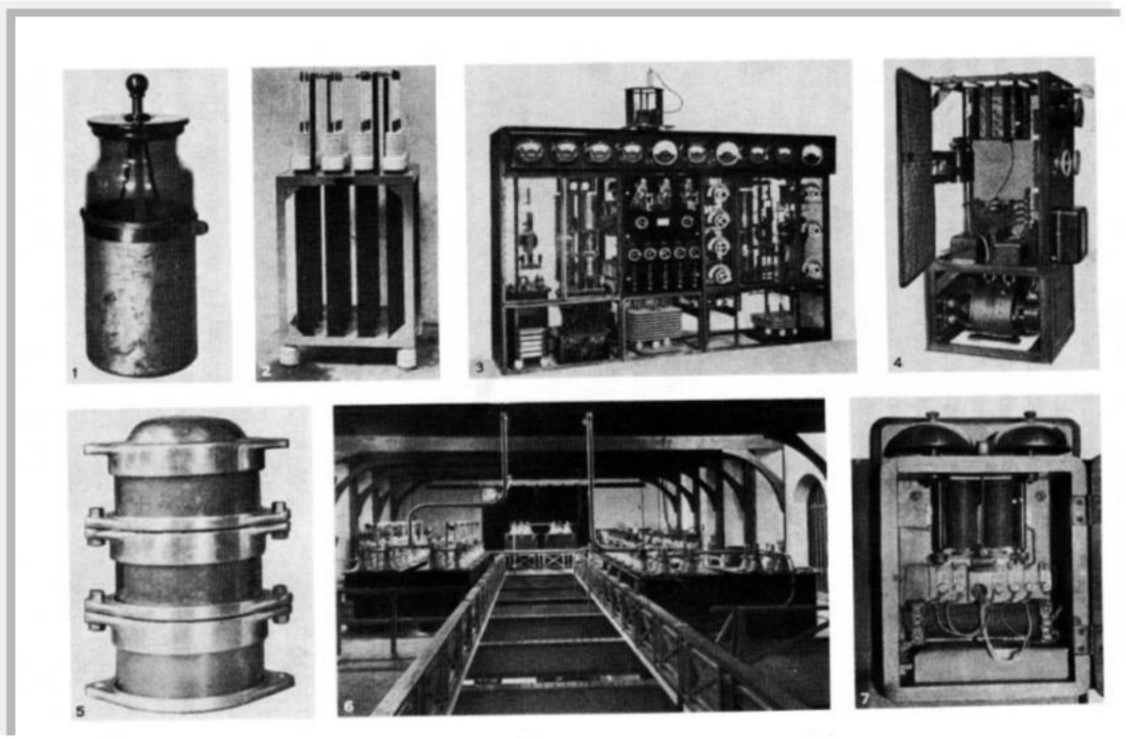


FIGURE 1.1. Photographs of various designs and arrangements of condensers (capacitor devices), including first the Leyden jar. (Reproduced from *Encyclopaedia Britannica*, 1929 edition, with permission). (1) A Leyden jar, the simplest type of condenser. (2) A battery of Moscicki glass condenser tubes—an improvement on the Leyden jar, introduced about 1904 by Moscicki. (3) The transmitter at one of the stations of the British Broadcasting Corp., showing two air dielectric condensers in the lower compartment (center and right-hand side). (4) Ship's radio transmitter, showing mica dielectric condenser on right side on shelf over generator. (5) Mica dielectric condenser of special construction for shortwave radio stations. (6) Condenser gallery of the Rugby radio station of the British post office. (7) Interior of the bell box used with British office telephone instrument, showing paper dielectric condenser (bottom of box). (8) Top row: several patterns of fixed condensers. Bottom row: several types of variable condensers commonly used as tuning capacitors in old radios.

不同类型的电容器

B. E. Conway, Electrochemical Supercapacitors——Scientific Fundamentals and Technologic Applications, Kluwer, 1999.pp. 4-5.

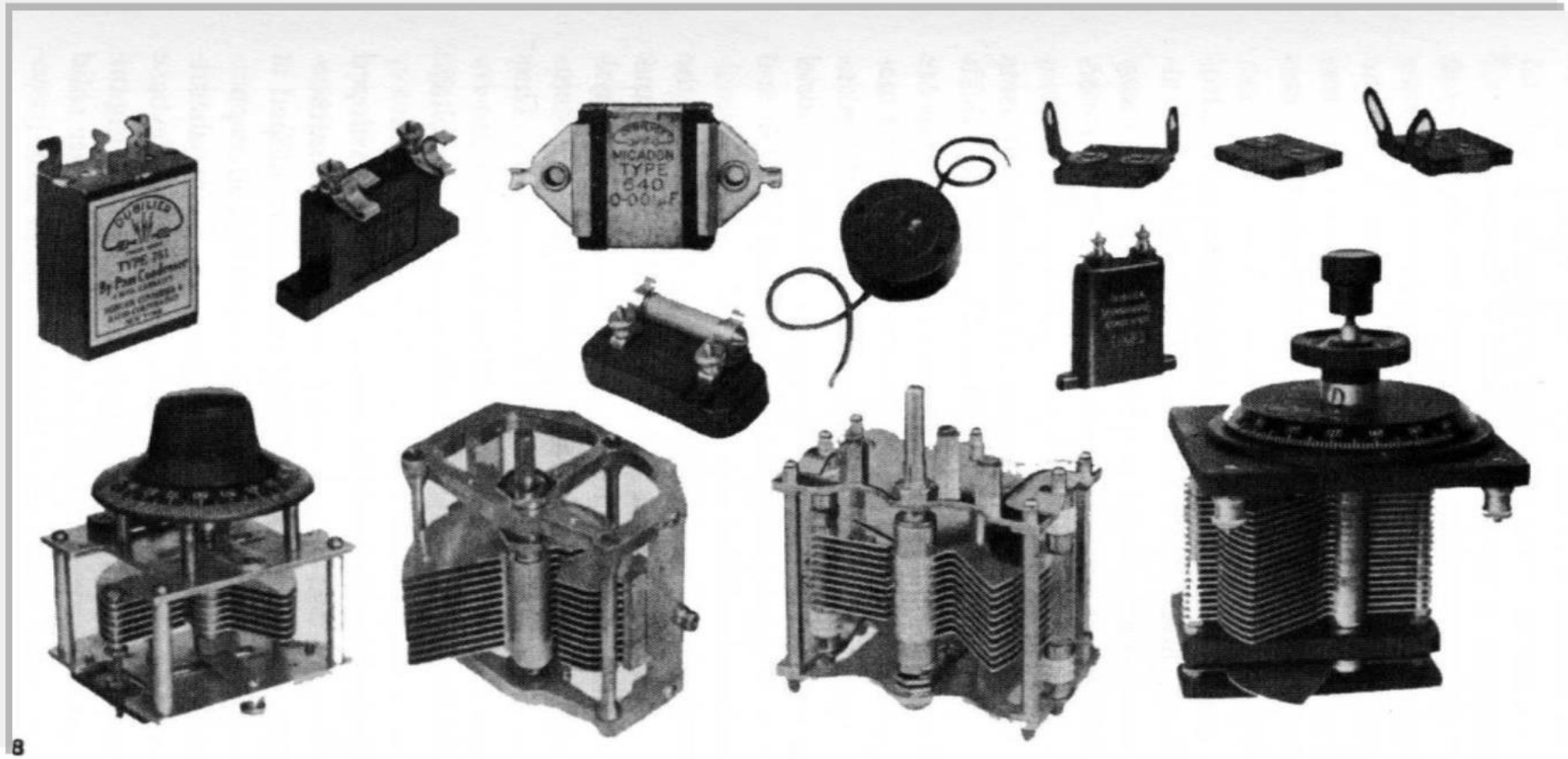
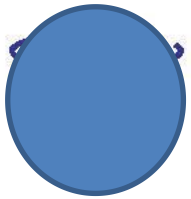


FIGURE 1.1. Photographs of various designs and arrangements of condensers (capacitor devices), including first the Leyden jar. (Reproduced from *Encyclopaedia Britannica*, 1929 edition, with permission). (1) A Leyden jar, the simplest type of condenser. (2) A battery of Moscicki glass condenser tubes—an improvement on the Leyden jar, introduced about 1904 by Moscicki. (3) The transmitter at one of the stations of the British Broadcasting Corp., showing two air dielectric condensers in the lower compartment (center and right-hand side). (4) Ship's radio transmitter, showing mica dielectric condenser on right side on shelf over generator. (5) Mica dielectric condenser of special construction for shortwave radio stations. (6) Condenser gallery of the Rugby radio station of the British post office. (7) Interior of the bell box used with British office telephone instrument, showing paper dielectric condenser (bottom of box). (8) Top row: several patterns of fixed condensers. Bottom row: several types of variable condensers commonly used as tuning capacitors in old radios.

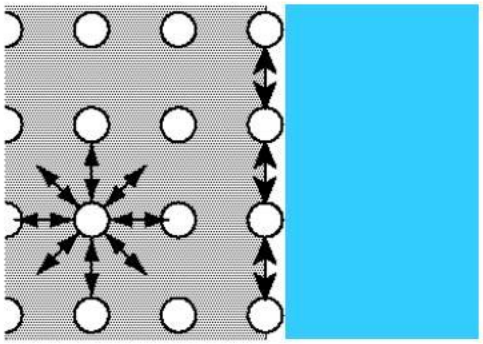
2.5 电极/溶液界面结构模型

金属胶体粒子



Zeta电位 Zeta电位仪

金属电极电解质界面



界面

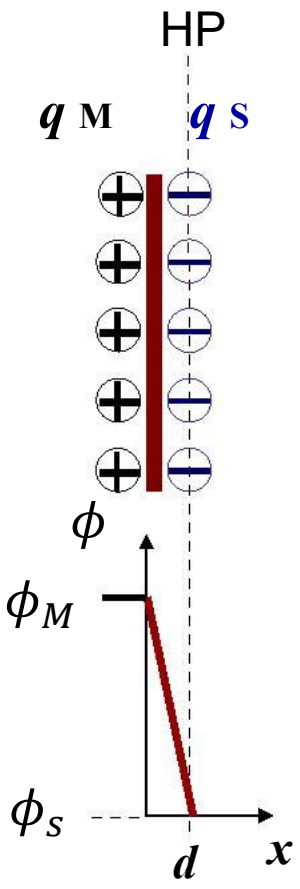
界面电位

电极电位 恒电位仪

微分电容 毛细管静电计

（1） 电极/溶液界面结构模型

Helmholtz 模型 (1853-1879) —紧密双电层模型



• 紧密层 (Compact layer)　$q_M + q_S = 0$

溶液一侧离子紧靠电极表面排列, 与表面仅发 生静电作用. 双电层厚度d ~ 离子半径.

• 类似于平板电容器

$$C_d = \frac{dq}{dE} = \frac{\varepsilon_r \varepsilon_0}{d} \sim 10\ \mu\mathrm{F\ cm^{-2}}$$

➢C_d 为常数, 不随所加电势而变化。

➢仅在浓溶液中, 且电势差较大时, 一定程度上符合实验结果.

□ 未考虑电解质浓度的影响, 稀溶液的计算值与实验结果不符合!

（1） 电极/溶液界面结构模型

- 平板电容器，无电活性物种

$$C_d = \frac{dq}{dE} = \frac{\varepsilon_r \varepsilon_0}{d} \sim 10 - 20\ \mu\text{F cm}^{-2}$$

$$1Farad = \frac{1coulomb \times 3 \times 10^9}{1volt \times \frac{1}{300}} = 9 \times 10^{11} esu$$

$$1\mu F = 10^{-6} F = 9 \times 10^5 esu$$

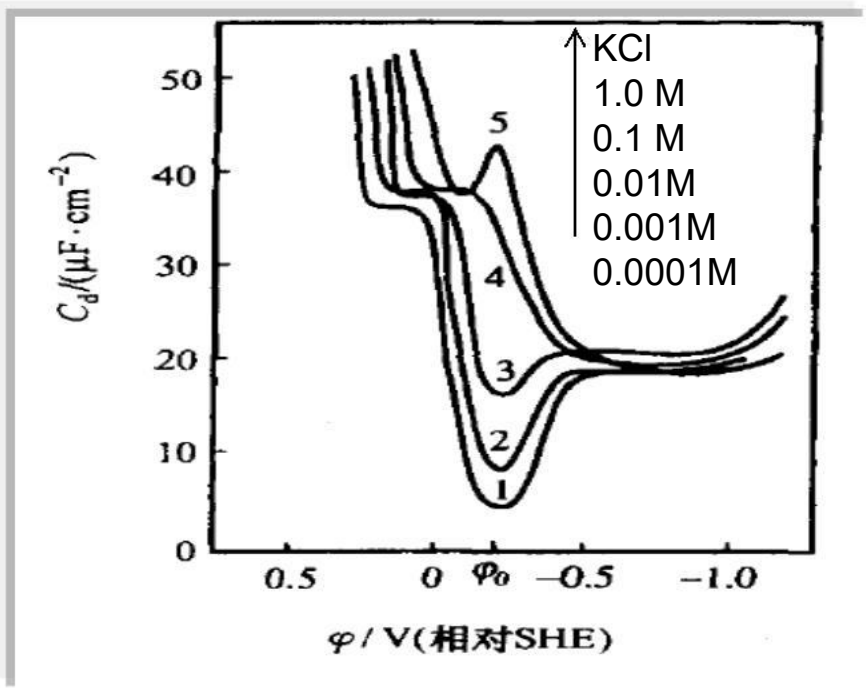
$$C_d = \frac{\varepsilon_r \varepsilon_0}{d} = \frac{\varepsilon_r \times 8.854 \times 10^{-12}\ F/m}{d(\text{m})} = 10 \sim 20 \mu F / cm^2$$

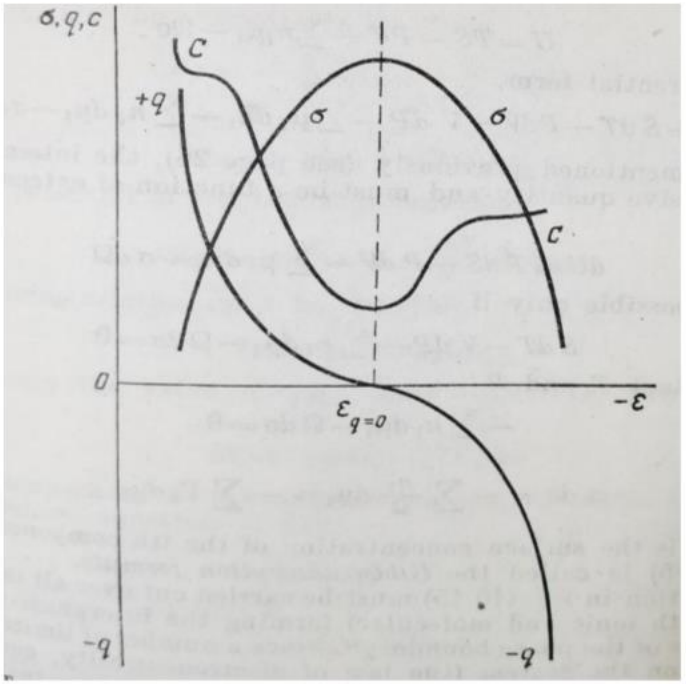
$$\frac{d}{\varepsilon_r} = \frac{\varepsilon_0}{C_d} = \frac{8.854 \times 10^{-12}\ F/m}{C_d} \simeq 1 \times 10^{-8} cm$$

实验测定电容随电位和电解质浓度变化

L. I. Antropov Electrochemical Principles, p297

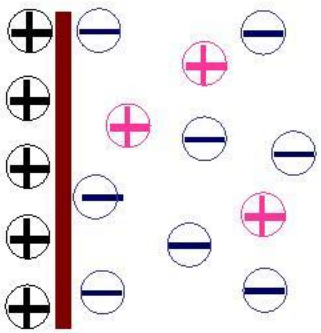
（1） 电极/溶液界面结构模型





L. I. Antropov Electrochemical Principles, p267

(2) Gouy-Chapman 模型(1910,1913)　(分散层模型)

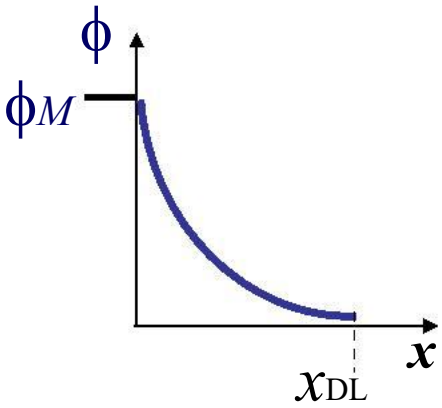


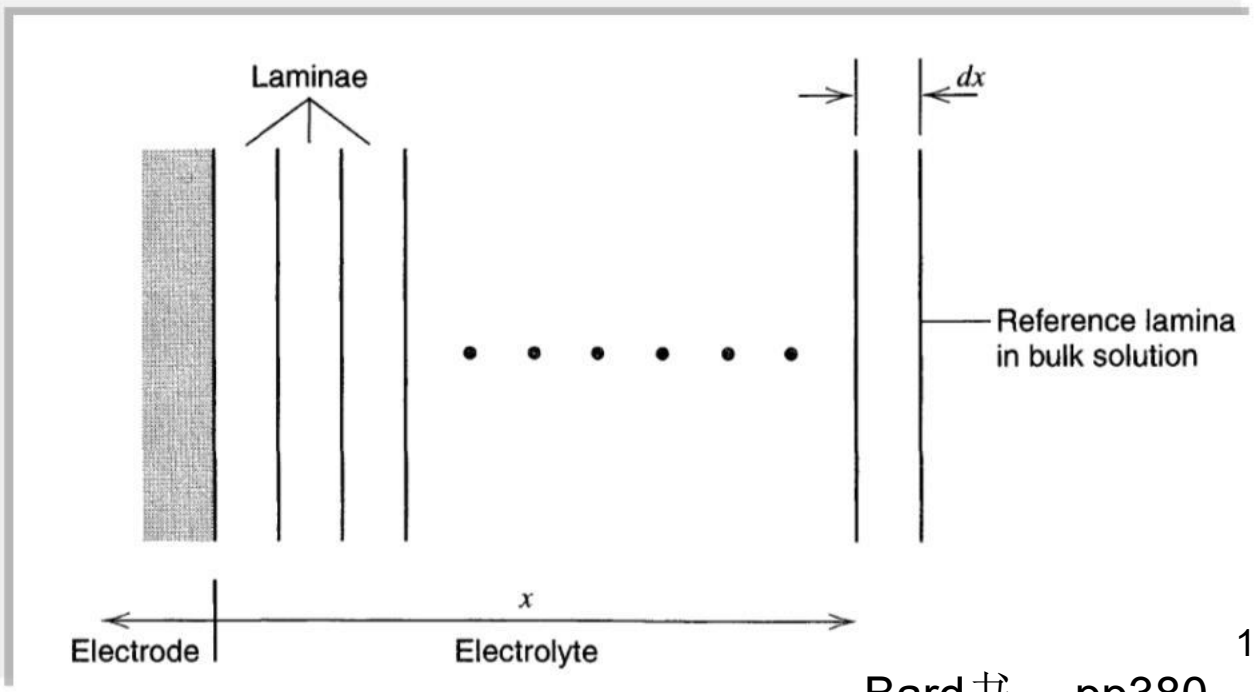
- 分散层 **(Diffuse layer)**

溶液一侧离子因静电作用和热运动而分散排列, 双电层可具有不同厚度.

- 定量描述分散层的离子分布与电位分布

假设及模型: **介质各向同性, 离子为点电荷, 静电作用**





Bard书， pp380

Poisson 方程　　静电场中电势与电荷分布的关系式

根据电场的散度定律

（微分形式的高斯定律）

$$\nabla \cdot E = \frac{1}{\varepsilon_r \varepsilon_0}\rho$$

$$\nabla = \frac{\partial}{\partial x}\vec{i} + \frac{\partial}{\partial y}\vec{j} + \frac{\partial}{\partial z}\vec{k}$$

Poisson 方程

$$\nabla^2 \phi = -\frac{1}{\varepsilon_r \varepsilon_0}\rho$$

Laplace算符

$$\nabla^2 = \nabla \cdot \nabla$$

矢量乘法运算：

点乘：$\vec{a} \cdot \vec{b}$?

叉乘：$\vec{a} \times \vec{b}$?

在直角坐标下（三维体系）

$$\nabla^2 = \frac{\partial^2}{\partial x^2} + \frac{\partial^2}{\partial y^2} + \frac{\partial^2}{\partial z^2}$$

$$\frac{\partial^2 \varphi}{\partial x^2} + \frac{\partial^2 \varphi}{\partial y^2} + \frac{\partial^2 \varphi}{\partial z^2} = -\frac{\rho}{\varepsilon_r \varepsilon_0}$$

一维体系：

（平板电极面）

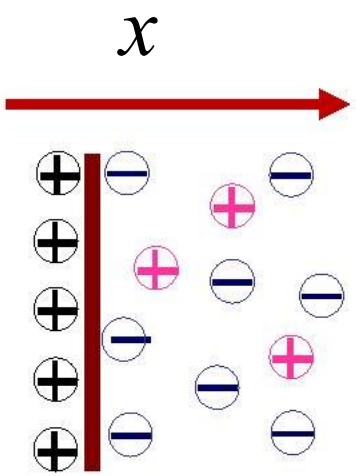
$$\frac{\partial^2 \varphi}{\partial x^2} = -\frac{\rho}{\varepsilon_r \varepsilon_0}$$

εr – 介质的相对介电常数

$\rho(x)$ – x点附近电量密度

（国际单位制）

Poisson-Boltzmann 方程（平面电极）



Poisson 方程　　– 静电场中电荷周围的电势分布

$$\frac{d^2\psi(x)}{dx^2} = -\frac{1}{\varepsilon_r \varepsilon_0} q(x)$$

ε_r – 介质的相对介电常数

$q(x)$ – x点附近电量密度

（国际单位制）

Boltzmann 分布　　–势能场中粒子的浓度分布

$$c_j(x) = c_j^0 \exp\left[-\frac{z_j F\,\psi(x)}{RT}\right]$$

$$\Rightarrow \quad q(x) = \sum_j z_j F\, c_j(x)$$

Poisson-Boltzmann 方程（平面电极）及其解

$$\frac{d^2\psi(x)}{dx^2} = -\frac{1}{\varepsilon_r \varepsilon_0} \sum_j z_j F\, c_j^0 \exp\left(-\frac{z_j F\,\psi(x)}{RT}\right)$$

(平板电极) Poisson-Boltzmann 方程的解（Bard书380页）

$$\frac{d^2\phi}{dx^2} = -\frac{1}{\varepsilon_r \varepsilon_0} \sum_j z_j F c_j^0 \exp\left(-\frac{z_j F \phi}{RT}\right)$$

1.变形
2.边界条件

(微分性质) ↓ $\frac{d^2\phi}{dx^2} = \frac{1}{2}\frac{d}{d\phi}\left\{\left(\frac{d\phi}{dx}\right)^2\right\}$

$$d\left(\frac{d\phi}{dx}\right)^2 = -\frac{2}{\varepsilon_r \varepsilon_0} \sum_j z_j F c_j^0 \exp\left(-\frac{z_j F \phi}{RT}\right) d\phi \longrightarrow d\left(\frac{d\phi}{dx}\right)^2 = \frac{2RT}{\varepsilon_r \varepsilon_0} \sum_j c_j^0 \, d\left[\exp\left(-\frac{z_j F \phi}{RT}\right)\right]$$

↓ 求积分

$$\left(\frac{d\phi}{dx}\right)_x^2 - \left(\frac{d\phi}{dx}\right)_\infty^2 = \frac{2RT}{\varepsilon_r \varepsilon_0} \sum_j c_j^0 \left[\exp\left(-\frac{z_j F \phi_x}{RT}\right) - \exp\left(-\frac{z_j F \phi_\infty}{RT}\right)\right]$$

↓ 利用边界条件 $x = \infty, \phi_\infty = 0, \left(\frac{d\phi}{dx}\right)_\infty = 0$ (求出常数)

$$\left(\frac{d\phi}{dx}\right)_x^2 = \frac{2RT}{\varepsilon_r \varepsilon_0} \sum_j c_j^0 \left[\exp\left(-\frac{z_j F \phi_x}{RT}\right) - 1\right]$$

（平面电极）**Poisson-Boltzmann** 方程的解

$$\left(\frac{d\phi}{dx}\right)_x^2 = \frac{2RT}{\varepsilon_r\varepsilon_0}\sum_j c_j^0\left[\exp\left(-\frac{z_jF\phi_x}{RT}\right)-1\right]$$

↓ 仅含一种z:z型电解质

$$\left(\frac{d\phi}{dx}\right)_x^2 = \frac{2RTc_0}{\varepsilon_r\varepsilon_0}\left[\exp\left(-\frac{zF\phi}{RT}\right)+\exp\left(\frac{zF\phi}{RT}\right)-2\right]$$

↓

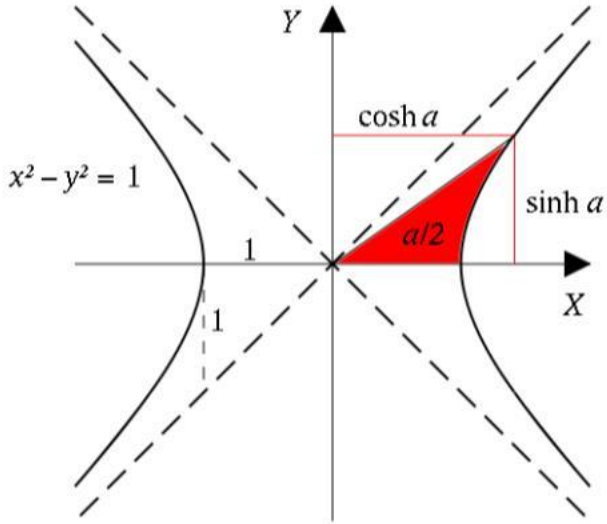
$$\left(\frac{d\phi}{dx}\right)_x^2 = \frac{2RTc_0}{\varepsilon_r\varepsilon_0}\left[\exp\left(\frac{zF\phi}{2RT}\right)-\exp\left(-\frac{zF\phi}{2RT}\right)\right]^2$$

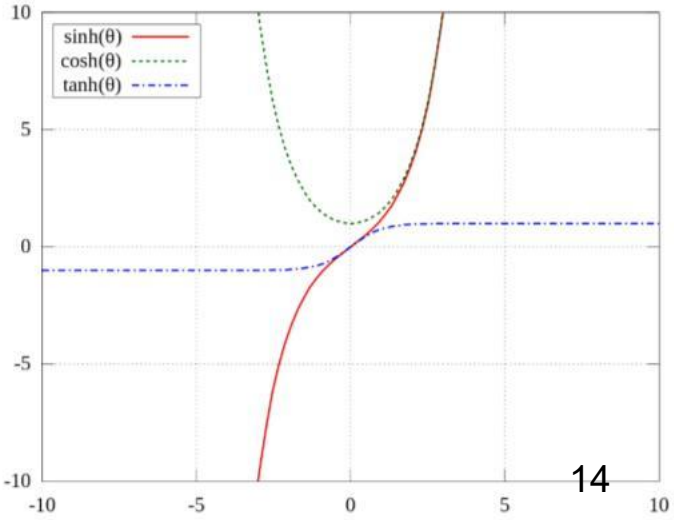
界面电场的场强 ↓

$$\frac{d\phi}{dx} = -\left(\frac{2RTc_0}{\varepsilon_r\varepsilon_0}\right)^{1/2}\left(e^{\frac{zF\phi}{2RT}}-e^{-\frac{zF\phi}{2RT}}\right)$$

$$= -\left(\frac{8RTc_0}{\varepsilon_r\varepsilon_0}\right)^{1/2}\sinh\left(\frac{zF\phi}{2RT}\right)$$

双曲正弦函数
$\sinh(x)=(e^x-e^{-x})/2$





（平面电极）**Poisson-Boltzmann** 方程的解

界面电势分布：

$$\int \frac{d\varphi}{\sinh\left(\frac{zF\varphi}{2RT}\right)} = \int \left(\frac{8RTC^0}{\varepsilon_r\varepsilon_0}\right)^{1/2} dx \qquad \frac{d\varphi}{dx} = \left(\frac{8RTC^0}{\varepsilon_r\varepsilon_0}\right)^{1/2} \sinh\left(\frac{zF\varphi}{2RT}\right)$$

$$\downarrow \quad y = \frac{zF\phi}{2RT}$$

$$\frac{4RT}{ZF}\int_{y_0}^{y} \frac{dy}{e^{y} - e^{-y}} = -\left(\frac{8RTc^0}{\varepsilon_r\varepsilon_0}\right)^{1/2} x$$

$$\downarrow$$

$$\frac{4RT}{ZF}\int_{y_0}^{y} \frac{e^{y}\,dy}{e^{2y} - 1} = -\left(\frac{8RTc^0}{\varepsilon_r\varepsilon_0}\right)^{1/2} x$$

微积分公式 ➡ $$\frac{2RT}{ZF}\ln\left[\frac{\tanh(y/2)}{\tanh(y_0/2)}\right] = -\left(\frac{8RTc^0}{\varepsilon_r\varepsilon_0}\right)^{1/2} x$$

（平面电极）**Poisson-Boltzmann** 方程的解

$$\frac{2RT}{zF}\ln\left[\frac{\tanh(y/2)}{\tanh(y_0/2)}\right]=-\left(\frac{8RTc^0}{\varepsilon_r\varepsilon_0}\right)^{1/2}x]$$

$$\downarrow \quad y=\frac{zF\phi}{2RT}$$

$$\ln\left[\frac{\tanh(zF\phi/4RT)}{\tanh(zF\phi_0/4RT)}\right]=-\left(\frac{2Z^2F^2C^0}{\varepsilon_r\varepsilon_0RT}\right)^{1/2}x]$$

$$\downarrow \quad \frac{\tanh(zF\phi/4RT)}{\tanh(zF\phi_0/4RT)}=e^{-\kappa x}$$

$$\downarrow \quad p=\ln\tanh(zF\phi_0/4RT)$$

$$\phi=\frac{4RT}{zF}\tanh^{-1}\exp(p-\kappa x)$$

浓度对双电层影响

$$\kappa=\left(\frac{2Z^2F^2C^0}{\varepsilon_r\varepsilon_0RT}\right)^{1/2}$$

分散层有效厚度

$$\frac{1}{\kappa}=\left(\frac{2Z^2F^2C^0}{\varepsilon_r\varepsilon_0RT}\right)^{-1/2}$$

$$\approx 0.096\,C_0^{-1/2}\ \text{Å}$$

浓度单位 mol/cm^3

分散层有效厚度

$$\kappa = \left(\frac{2n^0 z^2 e^2}{\varepsilon\varepsilon_0 kT}\right)^{1/2} = \left(\frac{2\frac{n^0}{N_A}N_A z^2 e^2}{\varepsilon\varepsilon_0 kT}\right)^{1/2} = \left(\frac{2c^0 N_A z^2 e^2}{\varepsilon\varepsilon_0 kT}\right)^{1/2}$$

对于稀水溶液 ε = 78.49，当 T = 25°C 时：

$$\kappa = \left(\frac{2C_0 \times 10^3 \times 6.02 \times 10^{23}\,\text{mol}^{-1} \cdot z^2 \times (1.6 \times 10^{-19}\,\text{C})^2}{78.49 \times 8.85 \times 10^{-12}\,\text{C}^2 \cdot \text{N}^{-1} \cdot \text{m}^{-2} \times 1.38 \times 10^{-23}\,\text{J} \cdot \text{K}^{-1} \times 298.15\,\text{K}}\right)^{1/2}$$

$$\approx 3.29 \times 10^9\, zC_0^{1/2}\,(\text{m}^{-1}) = 3.29 \times 10^7\, zc_0^{1/2}\,(\text{cm}^{-1})$$

注：C_0 的单位为 mol/L

当溶液为 1:1 型电解质水溶液，且 $c_0 = 0.1\,\text{mol} \cdot \text{L}^{-1}$ 时：

$$\kappa = 3.29 \times 10^7 \times 1 \times 0.1^{1/2}\,(\text{cm}^{-1}) = 1.04 \times 10^7\,(\text{cm}^{-1})]$$

$\Longrightarrow$ 分散层厚度 $L = \frac{1}{\kappa} = \frac{1}{1.04 \times 10^7}\,\text{cm} = 9.6 \times 10^{-8}\,\text{cm} = 0.96\,\text{nm}]$

分散层厚度, 即*Debye Length*

$$x_{DL} = \frac{1}{\kappa} = \left(\frac{\varepsilon_r \varepsilon_0 RT}{2c^0 z^2 F^2}\right)^{1/2}$$

ε_r 水的相对介电常数 78.4

ε_0 真空介电常数

$\varepsilon_0 = 8.85419 \times 10^{-12}\ \mathrm{F}/m$

T 绝对温度，T=298.15K

c^0 电解质浓度，单位mol/L

z 电解质离子的价态

F 法拉第常数

浓度与分散层厚度的关系

C^0 (mol/L)	*DL* (Å)
1	3.0
0.1	9.6
0.01	30.4
1×10^{-3}	96.2
1×10^{-4}	304

(25°C, I-I价电解质水溶液, c_0 为体相浓度)

$$F = 1.602176 \times 10^{-19}\ \mathrm{C}/e \times 6.02 \times 10^{23}\ e/mol = 96451\ \mathrm{C}/mol \approx 96500\ \mathrm{C}/mol$$

Poisson-Boltzmann 分布

分散层的电势分布

$$\frac{\tanh(zF\phi/4RT)}{\tanh(zF\phi_0/4RT)} = e^{-\kappa x}$$

ϕ_0 是在 x=0 处分散层的电势降，相对于溶液的电势。

$$\tanh(zF\phi/4RT) = \frac{\sinh(zF\phi/4RT)}{\cosh(zF\phi/4RT)}$$

$$= \frac{\exp(zF\phi/4RT) - \exp(-zF\phi/4RT)}{\exp(zF\phi/4RT) + \exp(-zF\phi/4RT)}$$

$$= \frac{\exp(zF\phi/2RT) - 1}{\exp(zF\phi/2RT) + 1}$$

$$\Longrightarrow \quad \tanh(zF\phi/4RT) \approx \frac{zF\phi}{4RT}$$

Poisson-Boltzmann 分布

分散层的电势分布

$$\frac{\tanh(zF\phi/4RT)}{\tanh(zF\phi_0/4RT)} = e^{-\kappa x}$$

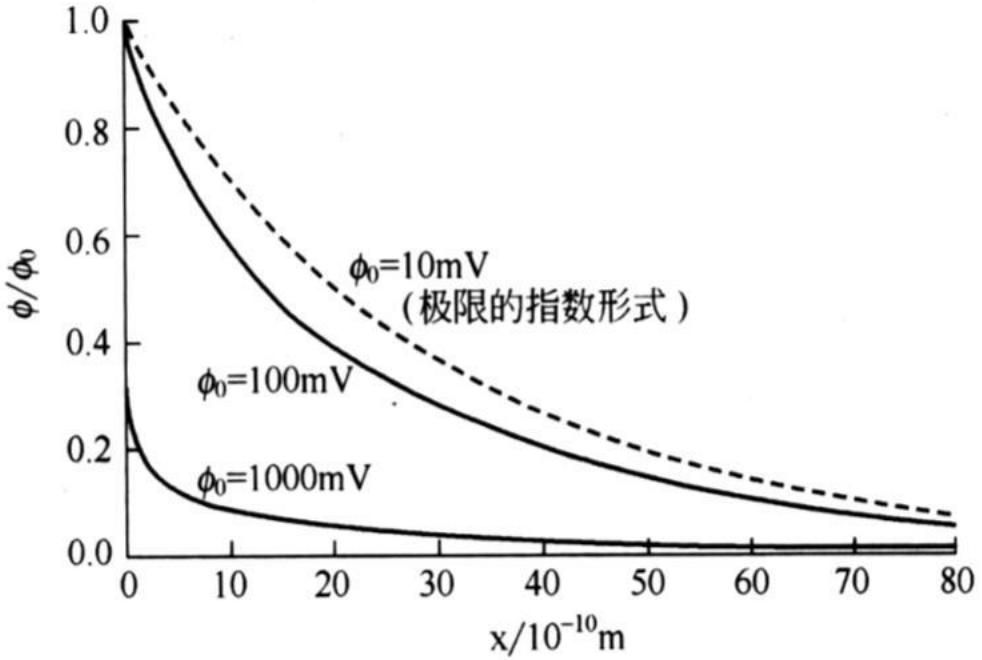
$$\tanh(zF\phi_0/4RT) \approx \frac{zF\phi_0}{4RT}$$

$$\phi_0 \leq \frac{50}{z}\,\text{mV} \quad (25^\circ C)$$

$$\frac{zF\phi_0}{2RT} \ll 2$$

$$\Longrightarrow \phi \approx \phi_0\, e^{-\kappa x}$$

Gouy-Chapman模型中分散层电势分布



➢ 浓度越大，分散层越薄，指数衰减越快.

➢ ϕ_0 越大，ϕ 衰减越快.

Debye-Hūckel理论 1923

离子氛Debye厚度

$z-z$ 价　　$x_{DL}=\frac{1}{\kappa}=\left(\frac{\varepsilon_r\varepsilon_0 RT}{2c^0 z^2 F^2}\right)^{1/2}$

z_i-z_j 价　　离子强度 $I=\frac{1}{2}\left(\sum_i c_i z_i^2\right)=\frac{1}{2}\left(\sum_i \nu_i z_i^2\right)m$

代替公式中的 $c^0 z^2$。

$$\frac{1}{\kappa}=\left(\frac{\varepsilon_0\varepsilon_r RT}{4F^2}\right)^{1/2}\cdot\frac{1}{\sqrt{I}}$$

Poisson-Boltzmann 方程（平面电极）及其解

$$\frac{d^2\psi(x)}{dx^2} = -\frac{1}{\varepsilon_r \varepsilon_0} \sum_j z_j F c_j^0 \exp\left(-\frac{z_j F \psi(x)}{RT}\right)$$

$$\frac{d\psi(x)}{dx} = -\left(\frac{8RTc_0}{\varepsilon_r \varepsilon_0}\right)^{1/2} \sinh\left(\frac{zF\phi}{2RT}\right)$$

$$\phi = \frac{4RT}{zF} \tanh^{-1}(p - \kappa x)$$

$$x_{DL} = \left(\frac{\varepsilon_r \varepsilon_0 RT}{2c^0 z^2 F^2}\right)^{1/2} = \frac{1}{\kappa}$$

$\sinh(x)=(e^x - e^{-x})/2$

$\cosh(x)=(e^x+e^{-x})/2$

$\tanh(x)=\sinh(x)/\cosh(x)$

$\coth(x)=1/\tanh(x)$

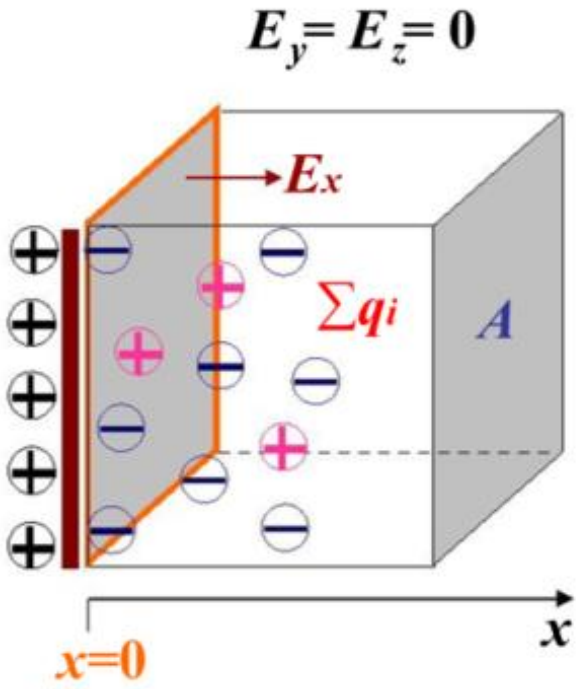
电极表面剩余电荷密度q_M与ϕ_0的关系

Gauss定理 $\oint \vec{E} \cdot d\vec{S} = \frac{1}{\varepsilon_r \varepsilon_0} \sum_i q_i$

$$q^S = \frac{\varepsilon_r \varepsilon_0}{A} \int_A \vec{E}_{x=0} \cdot d\vec{S} = \frac{\varepsilon_r \varepsilon_0}{A} \int_A E_{x=0}\, dS\cos\theta$$

$$= \frac{\varepsilon_r \varepsilon_0}{A} \int_A -E_{x=0} dS = -\varepsilon_r \varepsilon_0 E_{x=0}$$

$\cos\pi = -1$



$$q^M = -q^S = (8\varepsilon_r \varepsilon_0 RTc^0)^{1/2} \sinh\left(\frac{zF\phi_0}{2RT}\right)$$

$$q^M = 11.7\sqrt{c^0}\sinh(19.5\phi_0 z)$$

(25 °C, c^0 为 mol/L, q_M 为μC/cm²)

$$\frac{d\phi}{dx} = -\left(\frac{8RTc_0}{\varepsilon_r \varepsilon_0}\right)^{1/2} \sinh\left(\frac{zF\phi}{2RT}\right)$$

（电场强度等于电势梯度的负值）

➢ 一定浓度下，ϕ_0仅与 电极表面荷电状况有关，已知ϕ_0或q^M 可计算 q^M或ϕ_0

界面电容 (Interfacial capacitance)

– 反映界面电性质的重要参数

$$C_i = \frac{q}{E - E_{pzc}} \qquad q^M = C_i(E - E_{pzc})$$

微分电容 (C_d, Differential capacitance)

- 指定电势下,在外界电势扰动下表面剩余电荷密度的微小变化与扰动电势的比值

$$C_d = \left(\frac{\partial q^M}{\partial E}\right) \qquad q^M = \int_{E_{pzc}}^{E} C_d \, dE$$

$C_d \sim E$ 关系反映了电极表面随电位的结构变化。

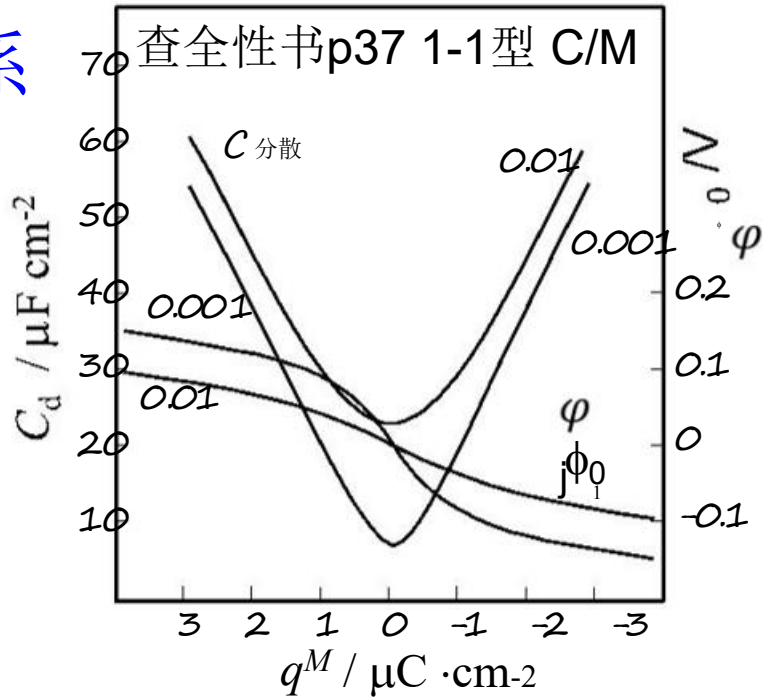
微分电容及其与浓度和电势的关系

$$q^M = -\,q^S = (8\varepsilon_r\varepsilon_0 RTc^{0\,1/2})\sinh\left(\frac{zF\phi_0}{2RT}\right)$$

$$C_d = \frac{dq^M}{d\phi_0} = \left(\frac{2z^2F^2\varepsilon_r\varepsilon_0 c_0^{1/2}}{RT}\right)\cosh\left(\frac{zF\phi_0}{2RT}\right)$$

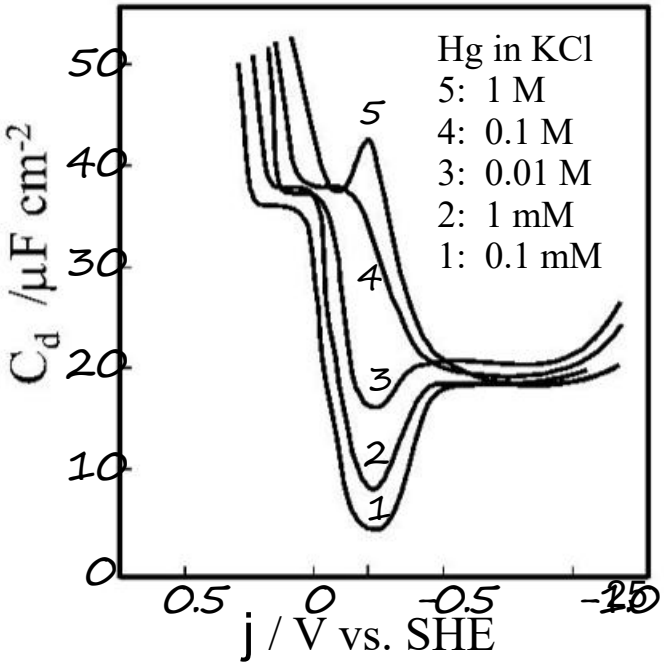
$$C_d = 228z\sqrt{c^0}\cosh(19.5\phi_0 z)$$

(25 ^{0}C, c^0 为 mol/L, C_d 为μF/cm^2)



➢稀溶液中, C_d在 q^M=0（即E_{PZC}）处有极小值, 而两边急剧增高, 在E_{PZC}附近符合实验趋势, 但比实验值高.

■认为离子是可无限接近表面的点电荷, 导致C_d无限增大, 不能解释浓溶液和偏离E_{PZC}情况下的实验结果。



13. $C_{2\text{-}S} = \frac{\partial \sigma_M}{\partial \phi_2} = 228.5|z|(C^b)^{1/2}\cosh(19.46|z|\phi_2)$

$C_{2\text{-}S}$ is given in $\mu F\,cm^{-2}$

$$\frac{1}{C} = \frac{1}{C_{M\text{-}2}} + \frac{1}{C_{2\text{-}S}}$$

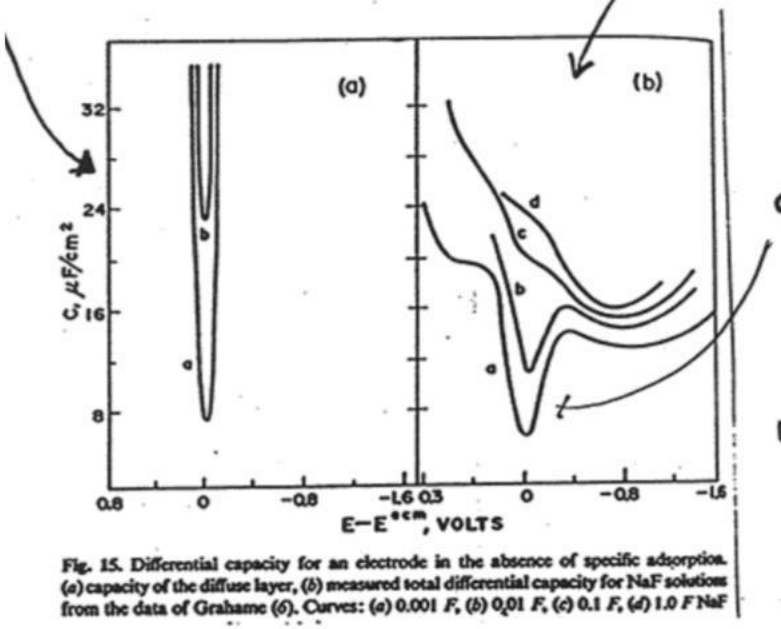
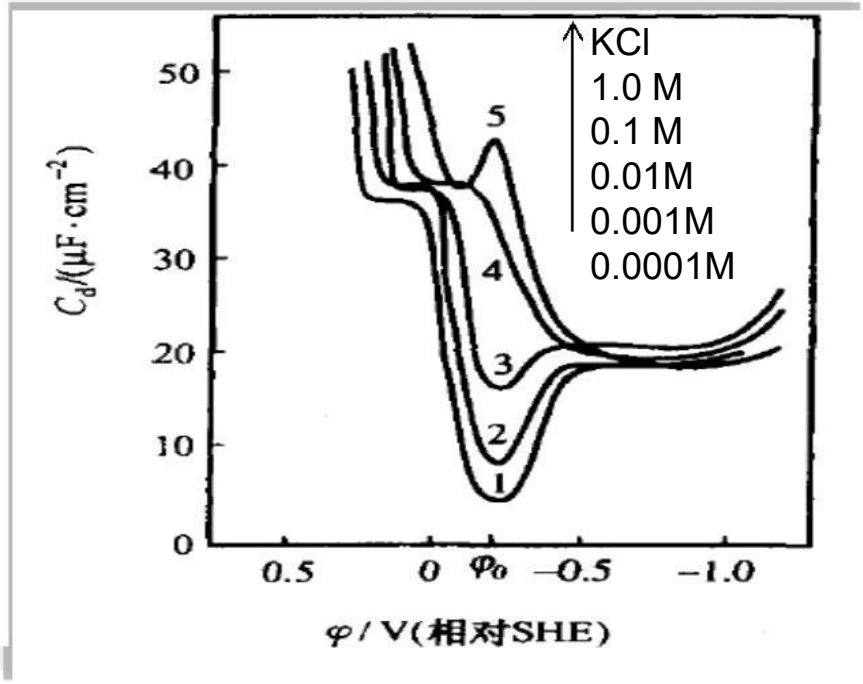
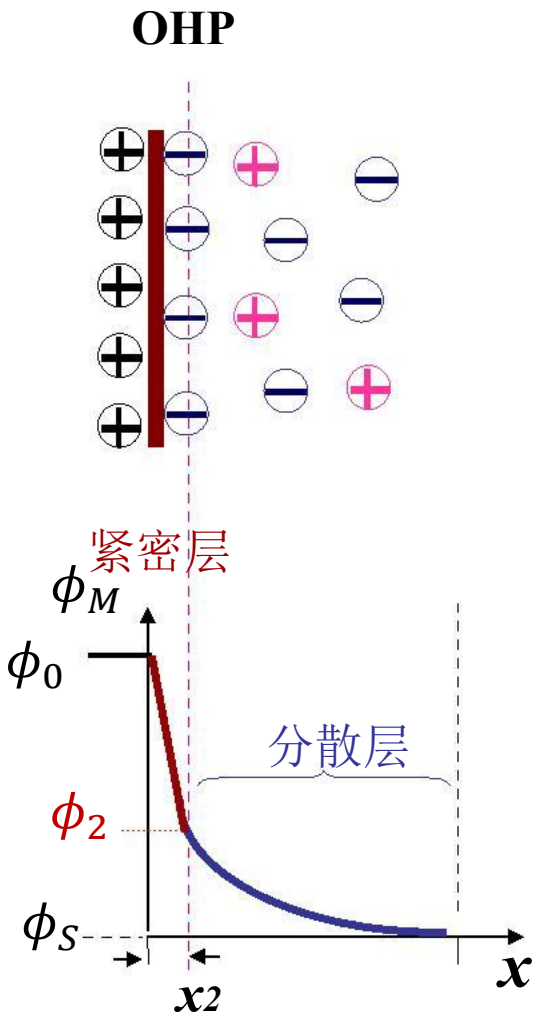


Fig. 15. Differential capacity for an electrode in the absence of specific adsorption. (a) capacity of the diffuse layer, (b) measured total differential capacity for NaF solutions from the data of Grahame (6). Curves: (a) 0.001 *F*, (b) 0.01 *F*, (c) 0.1 *F*, (d) 1.0 *F* NaF

diffuse layer minimum — its position is used to determine pzc



(3) Gouy-Chapman-Stern 模型（1924）
(GCS分散层模型，水溶液中双电层经典模型)



- **Stern** 对**Gouy-Chapman** 模型的修正
- 溶液一侧离子最靠近电极表面的距离为***d***, 所在位置称为OHP面.
- 水化离子

OHP (Outer Helmholtz Plane)
-最接近电极表面且与电极仅有静电相互作用的活性粒子(水化离子)中心面

- $x < x_2$ 内, 无电荷存在- 紧密层 (Helmholtz层);
  $x > x_2$ 的区间内, 离子因静电作用和热运动而分散排列-分散层
  Gouy-Chapman模型成立

1. GCS分散层特征参数与x的关系

电势分布

$$\phi(x)=\frac{4RT}{zF}tanh^{-1}\left[tanh\left(\frac{zF\phi_2}{4RT}\right)e^{-\kappa(x-x_2)}\right]$$

$(x\geq x_2,\phi_2$为x_2处的电势$)$

表面剩余电荷

$$q^M=\sqrt{8\varepsilon_r\varepsilon_0RTc^0}\,sinh\left(\frac{zF\phi_2}{2RT}\right)$$

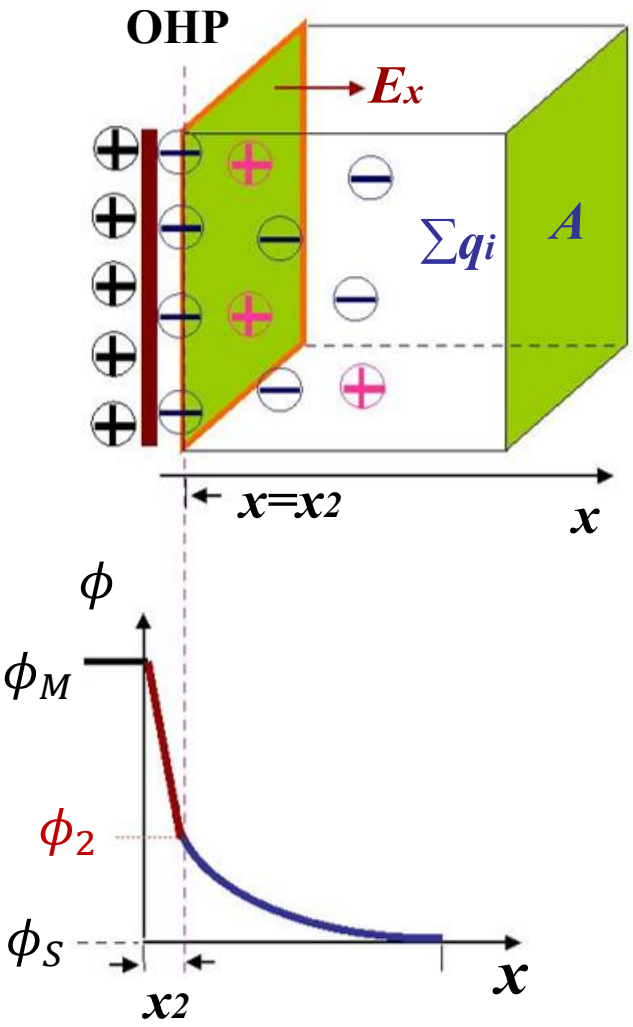
分散层电容

$$C_d=\frac{dq^M}{d\phi_2}=\left(\frac{2z^2F^2\varepsilon_r\varepsilon_0c^0}{RT}\right)^{1/2}cosh\left(\frac{zF\phi_2}{2RT}\right)$$

当$|\phi_2|\ll 50/zmV$时，$\phi(x)\approx\phi_2\cdot e^{-\kappa(x-x_2)}$

$$q^M\approx\sqrt{\frac{2\varepsilon_r\varepsilon_0c^0}{RT}}zF\phi_2$$

$$C_d\approx(2\varepsilon_r\varepsilon_0c^0/RT)^{1/2}zF$$



OHP处 电势变化斜率连续

$$\left.\frac{d\phi}{dx}\right|_{x\geq x_2}=-\left(\frac{8RTc_0}{\varepsilon_r\varepsilon_0}\right)^{1/2}sinh\left(\frac{zF\phi_2}{2RT}\right)$$

$$\approx-1.44\times10^5q^M(Vcm^{-1})$$

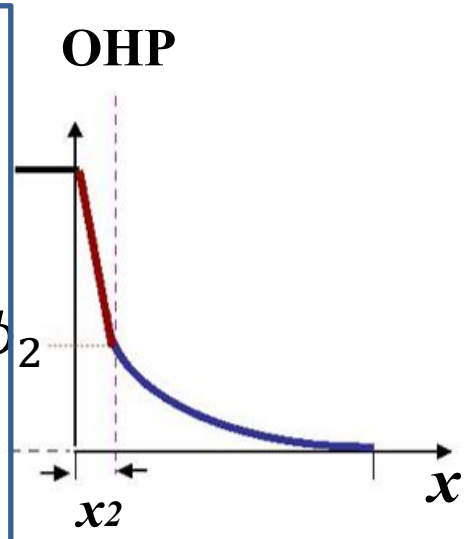
ϕ_2电势与相间电势差$\phi_M - \phi_s$的关系

$$\phi_M - \phi_2 = \left(\frac{d\phi}{dx}\right)\Big|_{x_2} x_2$$ 后面一项对应紧密层电位降

$$= \left(\frac{8RTc_0}{\varepsilon_r\varepsilon_0}\right)^{1/2} sinh\left(\frac{zF\phi_2}{2RT}\right) \cdot x_2$$

$$= \frac{1}{C_H}\sqrt{\frac{\varepsilon_r\varepsilon_0}{8RTc_0}} sinh\left(\frac{zF\phi_2}{2RT}\right)$$

$$= \frac{1}{C_H}\sqrt{2\varepsilon_r\varepsilon_0 RTc^0}\left[exp\left(\frac{zF\phi_2}{2RT}\right) - exp\left(-\frac{zF\phi_2}{2RT}\right)\right]$$



- 剩余电荷密度q^M和c^0都很小时

$$q^M = \sqrt{8\varepsilon_r\varepsilon_0 RTc^0} sinh\left(\frac{zF\phi_2}{2RT}\right)$$

ϕ_2值小，$(zF\phi_2/2RT)$小

$$\phi_M - \phi_2 \approx \phi_2$$

$$\frac{d\phi}{dx}\Big|_{x=x_2} = -\left(\frac{8RTc_0}{\varepsilon_r\varepsilon_0}\right)^{1/2} sinh\left(\frac{zF\phi_2}{2RT}\right)$$

➢ 相间电势主要分布在分散层中 (OHP处的电势变化率较小)

$$\phi_M - \phi_2 = \frac{1}{C_H}\sqrt{8\varepsilon_r\varepsilon_0 RTc^0}\left[exp\left(\frac{zF\phi_2}{2RT}\right) - exp\left(-\frac{zF\phi_2}{2RT}\right)\right]$$

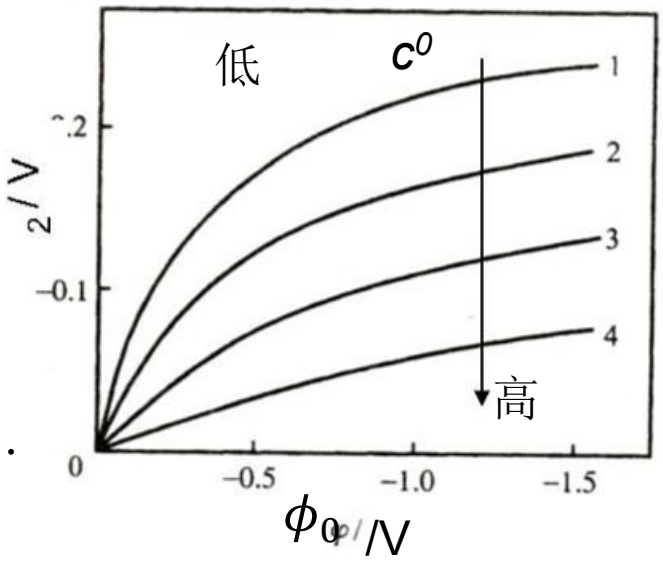
- 剩余电荷密度q_M很大时 (远离pzc),

➢ 相间电势主要分布在紧密层, 液相中的剩余电荷主要分布在OHP 处

$$\phi_M - \phi_2 \approx \pm\frac{1}{C_H}\sqrt{\frac{\varepsilon_r\varepsilon_0 RTc^0}{8}}exp\left(\pm\frac{zF\phi_2}{2RT}\right)$$

$$\phi_2 \approx \pm ln(\pm\phi_M) \mp \frac{RT}{zF}ln\, c_0 \mp \frac{2RT}{zF}ln\frac{1}{C_H}\sqrt{\frac{\varepsilon_r\varepsilon_0}{2RT}}$$

$$\left(\frac{\partial\phi_2}{\partial lg\, c_0}\right)_{\phi_M} \approx \mp\frac{2.303RT}{zF}$$



➢$|\phi_M|$增大, $|\phi_2|$随之增大, 但增加幅度较$|\phi_M|$小得多.

➢恒定ϕ_M, 浓度每增加10倍, $|\phi_2|$减小 59/z mV.

□ GCS 模型未考虑紧密层的结构细节, 如离子特性吸附和溶剂偶极取向, 仍然不能解释 ϕ_M远离 ϕ_{pzc}时的实验结果.

分散层中指定离子剩余浓度的表示式

$$c_j(x) = c_j^0 exp[-z_j F\phi(x)/RT]$$

$$\Gamma_{j,S} = \int_{x_2}^{\infty} (c_j(x) - c_j^0)dx$$

$$= c_j^0 \int_{x_2}^{\infty} (exp[-z_j F\phi(x)/RT] - 1)dx$$

$$\frac{tanh(zF\phi/4RT)}{tanh(zF\phi_2/4RT)} = e^{-(x-x_2)/x_{DL}}$$

$z-z$价型

$$\Gamma_{-,S} = \frac{\sqrt{2RT\varepsilon c^0}}{zF}$$

$$\varepsilon = \varepsilon_r \varepsilon_0$$

$$q_{-S} = -zF\Gamma_{-S} \frac{\{exp[zF\phi_2/2RT] - 1\}}{\sqrt{RT\varepsilon\varepsilon_0}\{exp[zF\phi_2/2RT] + 1\}}$$

$$\Gamma_{-,S} = \frac{\sqrt{2RT\varepsilon c^0}}{zF}\{exp[-zF\phi_2/2RT] - 1\}$$

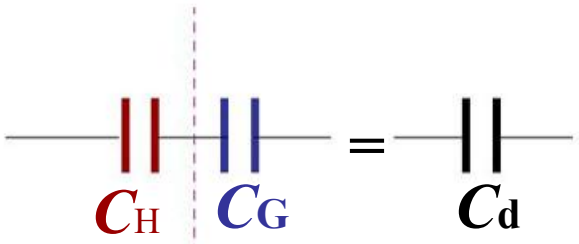
$$q_{-S} = zF\Gamma_{-,S} = \sqrt{2RT\varepsilon c_0}\{exp[-zF\phi_2/2RT] - 1\}$$

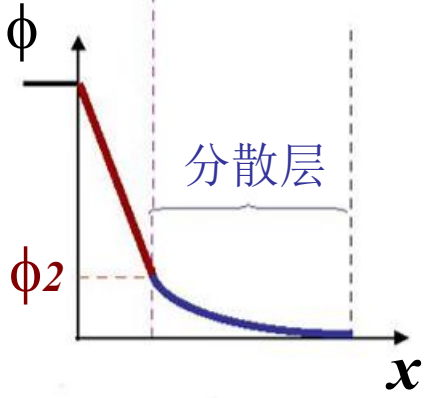
$$q_S = \sum_j z_j F\Gamma_{j,S} = -\sqrt{8RT\varepsilon c^0}\, sinh(zF\phi_2/2RT)$$

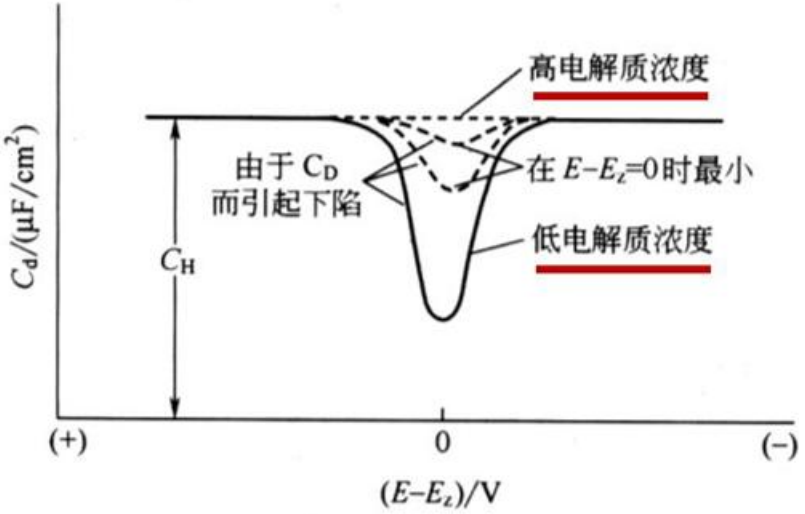
$$q^M = -q^S = \sqrt{8RT\varepsilon c^0}\, sinh(zF\phi_2/2RT)$$

公式中的z 不带符号

微分电容随电极电势和浓度的变化







$$\frac{1}{C_d}=\frac{1}{C_H}+\frac{1}{C_G}$$

$$C_d=\frac{1}{\frac{d}{d\phi_2}\left[\frac{1}{\varepsilon_r\varepsilon_0}\frac{(2z^2F^2\varepsilon_r\varepsilon_0c_0)^{1/2}}{RT}cosh\left(\frac{zF\phi_2}{2RT}\right)\right]}$$

- $E \to E_{pzc}$(即$\phi_2 \to 0$)时，$C_G \to$ 极小，

  $C_G \ll C_H$，　$C_d \approx C_G = \left(\frac{2z^2F^2\varepsilon_r\varepsilon_0c^0}{RT}\right)^{1/2}$

  浓度增加，x_{DL}减小，C_G 增大，$\boldsymbol{C_d}$增大

- $|E-E_{pzc}| \gg 0(|\phi_2| \gg 0)$时，

  $C_G >> C_H$，且高浓度下分散层电容减小，为紧密层结构，　$C_d \approx C_H$

➢ 能较好地解释$|\phi| \sim \phi_{pzc}$时C_d 实验结果及随浓度变化.

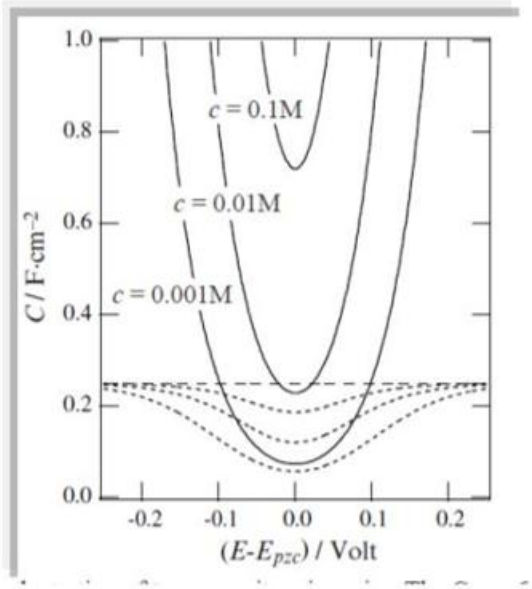
微分电容随电极电势和浓度的变化

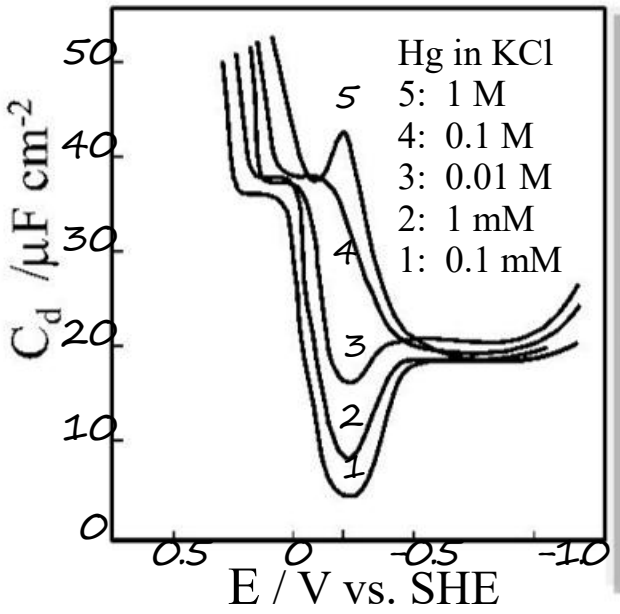
$$\frac{1}{C_d}=\frac{1}{C_H}+\frac{1}{C_G}$$

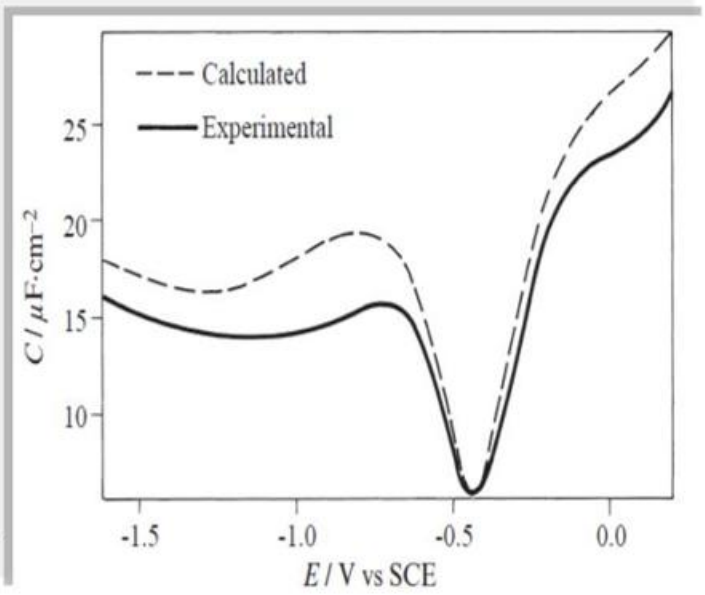
$$C_G=\frac{1}{\frac{d}{d\phi_2}\left[\frac{1}{\varepsilon_r\varepsilon_0}\frac{(2z^2F^2\varepsilon_r\varepsilon_0C^0)^{1/2}}{RT}cosh\left(\frac{zF\phi_2}{2RT}\right)\right]}$$

$$C_d\approx C_G=\left(\frac{2z^2F^2\varepsilon_r\varepsilon_0C^0}{RT}\right)^{1/2}cosh\left(\frac{zF\phi_2}{2RT}\right)$$

$$C_{d,pzc}\approx C_{G,pzc}=\left(\frac{2z^2F^2\varepsilon_r\varepsilon_0C^0}{RT}\right)^{1/2}$$







$$C_{紧}=\frac{\varepsilon_r\varepsilon_0}{d}\sim 10\mu Fcm^{-2}$$

$C_{紧}$: 高浓度微分电容的实验值

无特性吸附时紧密层电容

表面剩余电荷

$$q^M = \sqrt{8\varepsilon_r\varepsilon_0 RTc^0}\,sinh\left(\frac{zF\phi_2}{2RT}\right)$$

计算分散层电容

$$C_G = \frac{dq^M}{d\phi_2} = \left(\frac{2z^2F^2\varepsilon_r\varepsilon_0}{RT}c^0\right)^{1/2} cosh\left(\frac{zF\phi_2}{2RT}\right)$$

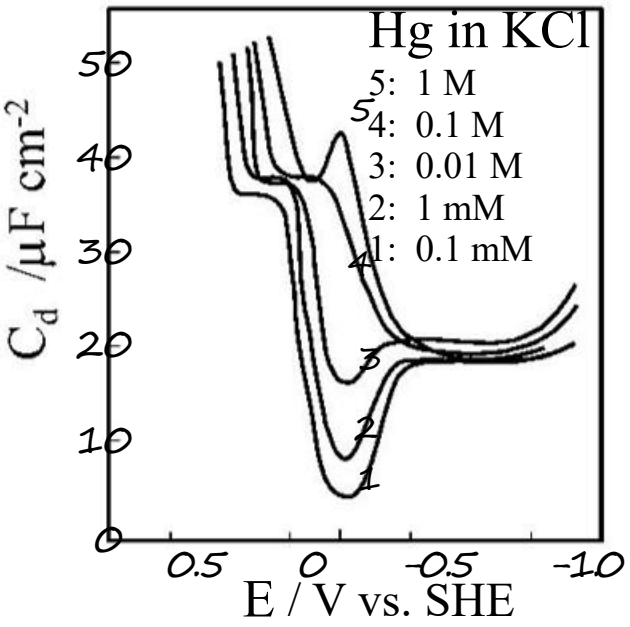
实验测得微分电容C_d

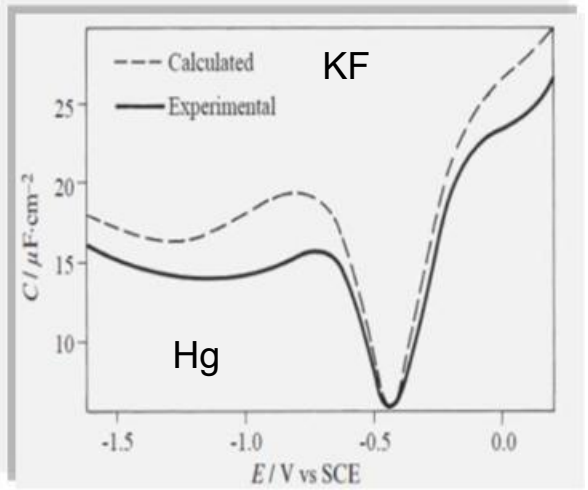
$$\frac{1}{C_d} = \frac{1}{C_H} + \frac{1}{C_G} \Rightarrow C_H$$

- $E \to E_{pzc}$(即$q \to 0, \phi_2 \to 0$)时，C_G → 极小，低浓度，$C_d \approx C_G$
- $|E - E_{pzc}| \gg 0 (q \gg 0, |\phi_2| \gg 0)$时，$C_G \gg C_H$，$C_d \approx C_H$;

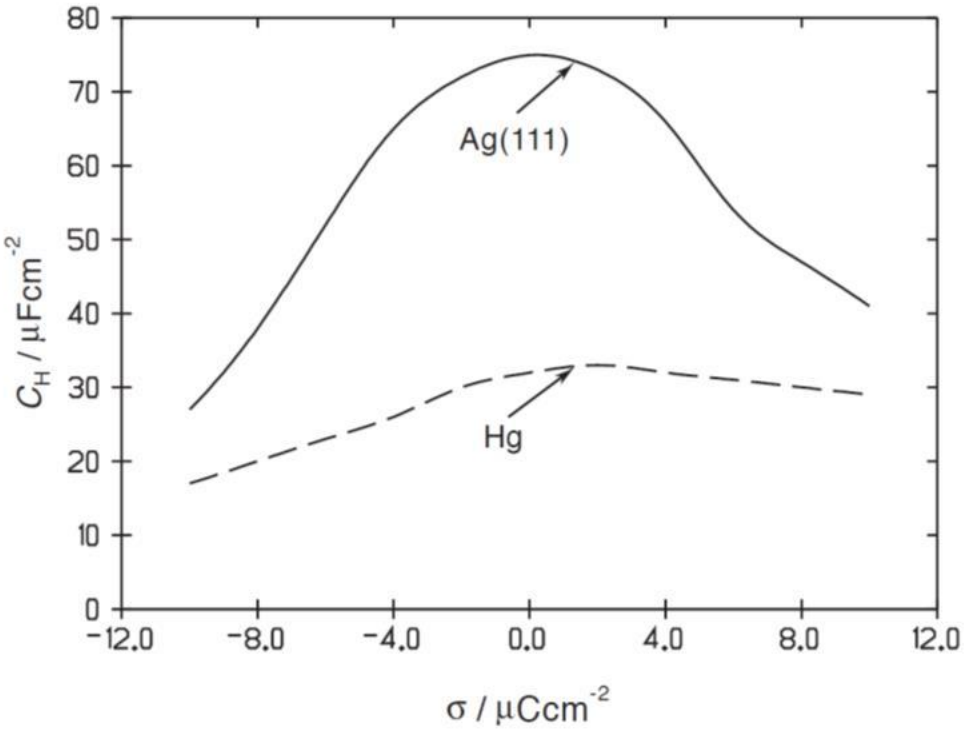
➢ 浓度不很稀时， 实验所得微分电容曲线可近似看成紧密层微分电容曲线。

➢ 紧密层电容由溶剂化层结构和离子水化半径决定。





➢ C_H ~ E



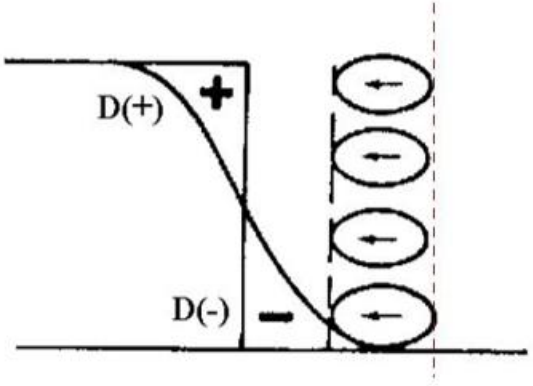
水溶液中Ag(111)和汞电极Helmholtz电容

W. Schmickler, Interfacial Electrochemistry, Springer-Verlag Berlin Heidelberg 2010. pp44 (图5.4）

金属对双电层内层电容的作用

金属表面电子密度分布-“电子气”模型，或 **Jellium**（冻胶）模型

$$\frac{1}{C_d}=\frac{1}{C_H}+\frac{1}{C_G}\,\frac{d}{\varepsilon_r\varepsilon_0}\,\frac{1}{\left(\frac{2z^2F^2\varepsilon_r\varepsilon_0C^0}{RT}\right)^{1/2}\cosh\left(\frac{zF\phi_2}{2RT}\right)}$$



0.1-0.2 nm

极性溶剂的分子取向

零电荷电势附近的紧密层电容为内层电容C_m 和溶液电容C_{sol} 之和：

$$(1/C_H)=(1/C_m)+(1/C_{sol})$$

➢ 在金属表面层中，电子密度呈现较快的衰减，造成势垒，自由电子溢出金属表面；

➢产生表面电势χ_M，金属/溶液界面区的总电位差应是金属项与溶液项之和；

➢ 电子溢出表面的距离 0.1-0.2 nm。

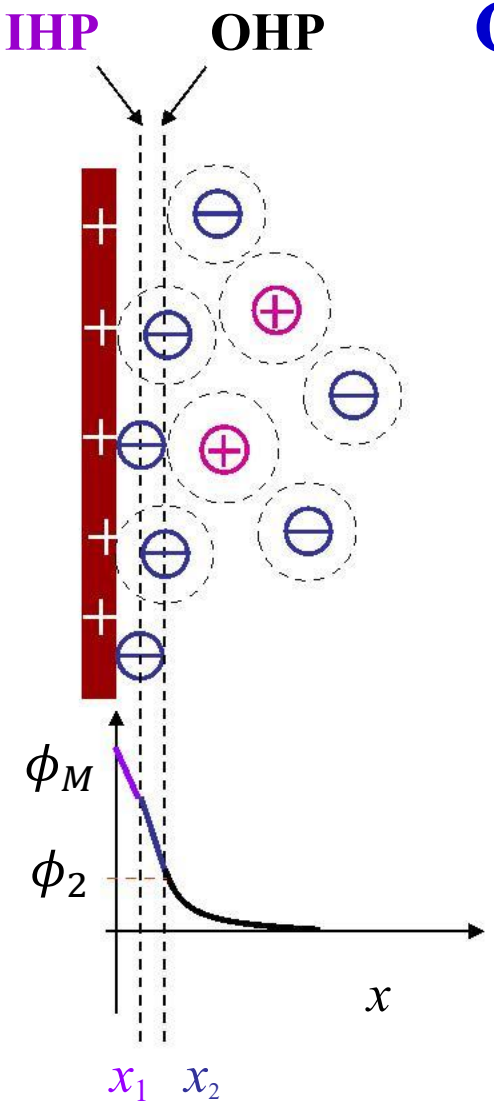
内层电容C_m 由表面电位χ随表面电荷密度q_M的变化所给定，

$$(1/C_m)=(\partial\chi/\partial q_M)$$

➢ C_m 具有负值， 故导致增大C_H 数值。

（吴辉煌教材 p16, 金属的作用）

（4）特性吸附时紧密层的结构



Grahame模型 (1947) - 离子特性吸附模型

- 离子特性吸附(Specific Adsorption)

- 去溶剂化离子紧密地与电极表面接触，除静电作用外还与电极表面发生短程作用；

IHP (Inner Helmholtz Plane, 内双层)

- 特性吸附离子层的中心面

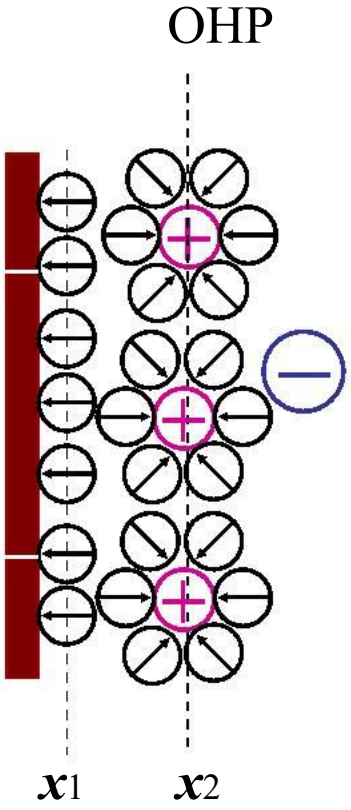
- 特性吸附离子可与电极带相同或相反的电荷，在原无特性吸附体系的*PZC*处也可吸附.

- 正离子的水化作用强，发生特性吸附的离子通常为阴离子；

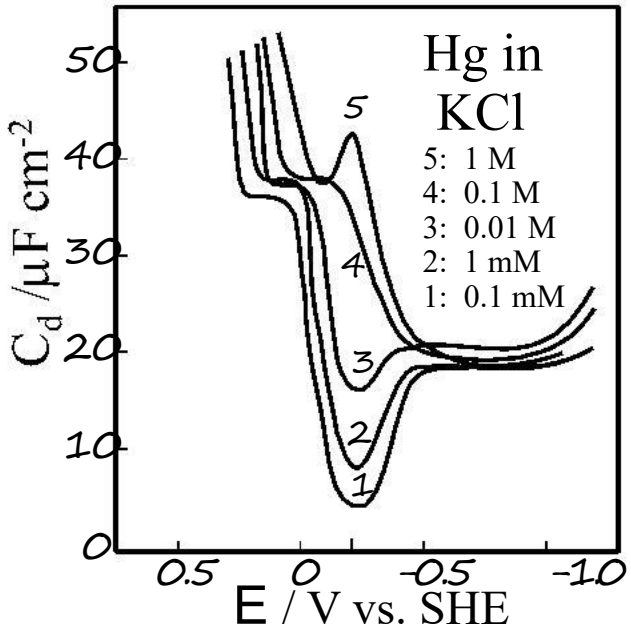
紧密层的结构

Bockris-Devanathan-Muller (BDM)模型 (1963)

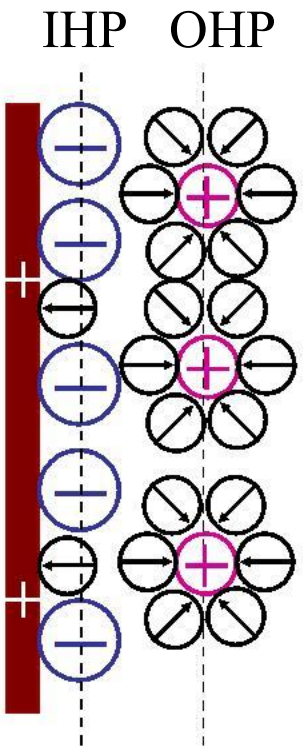
- 溶剂层模型



无特性吸附

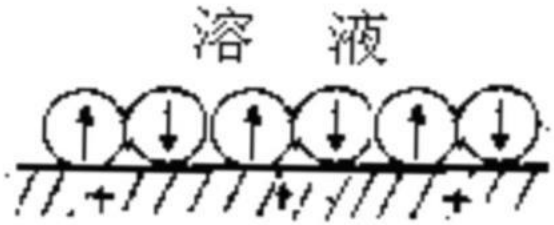


•表面极性溶剂分子介电饱和，无特性吸附时，水分子层的电容决定紧密层的电容,微分电容保持恒定.



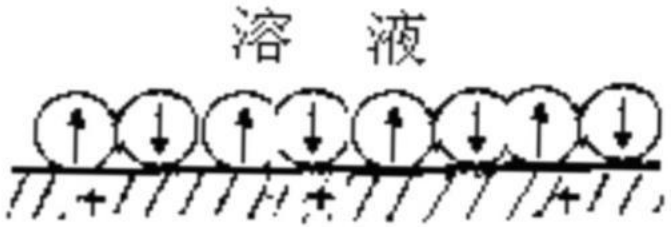
有特性吸附

溶剂偶极取向



Watts-Tobin 模型

二态模型，水分子以单体存在



Bockris-Habib 模型

三态模型（偶极向上和向下的水单体，加上没有静偶极矩的二聚体）



Damaskin- Frumkin 模型

四态模型， 水单体，加上偶极向上和向下的水分子聚集体

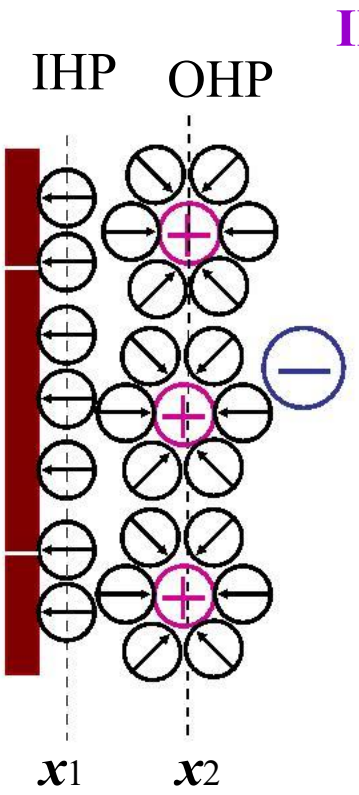


Parsons 模型

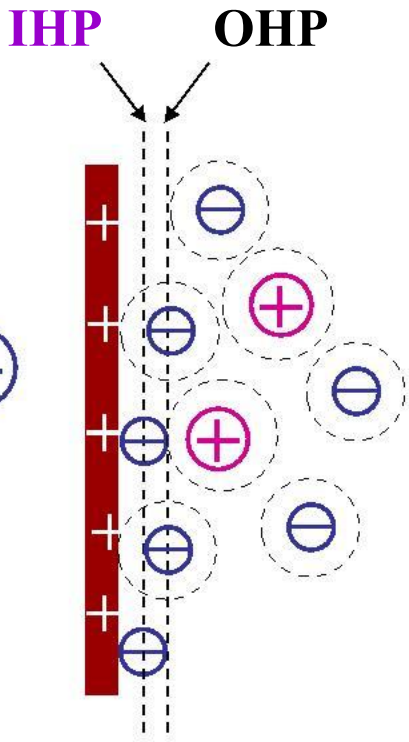
四态模型， 偶极向上和向下的水单体，加上偶极向上和向下的水分子聚集体

电偶极矩（ μ ）由负电荷指向正电荷

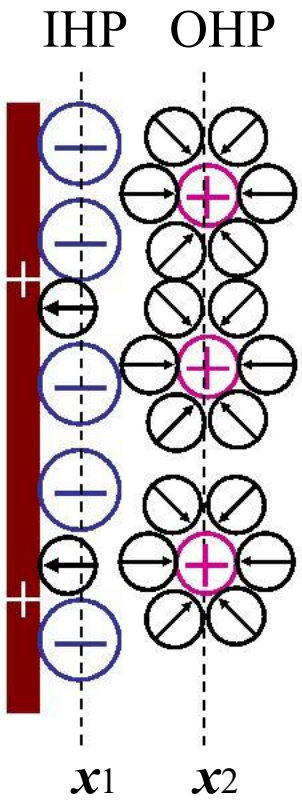
几种紧密层的结构和IHP的定义



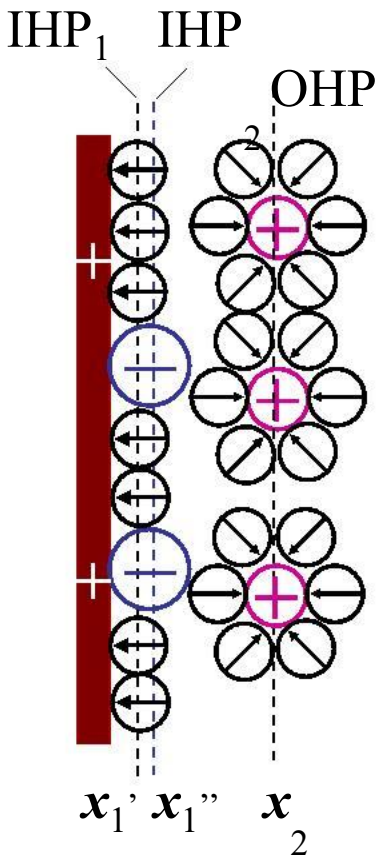
无特性吸附



特性吸附



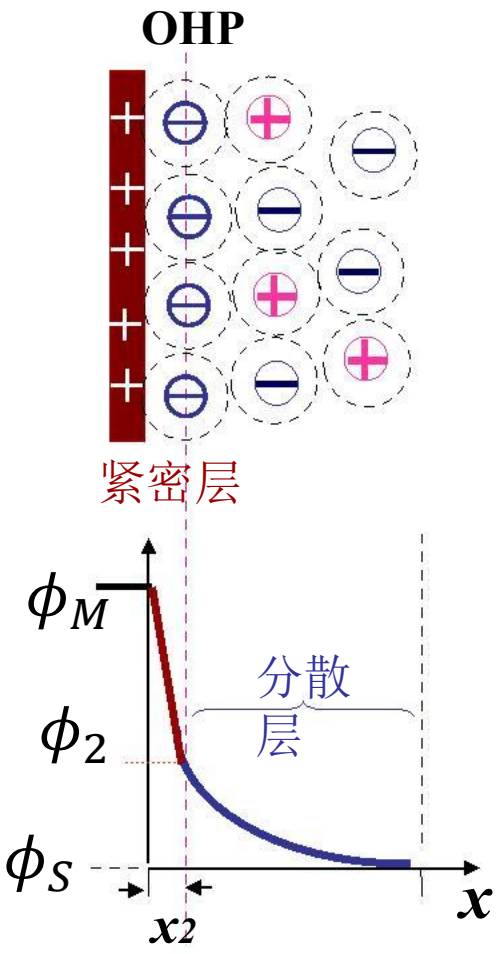
特性吸附(超载)



有特性吸附

有分子偶极层

GCS分散层模型

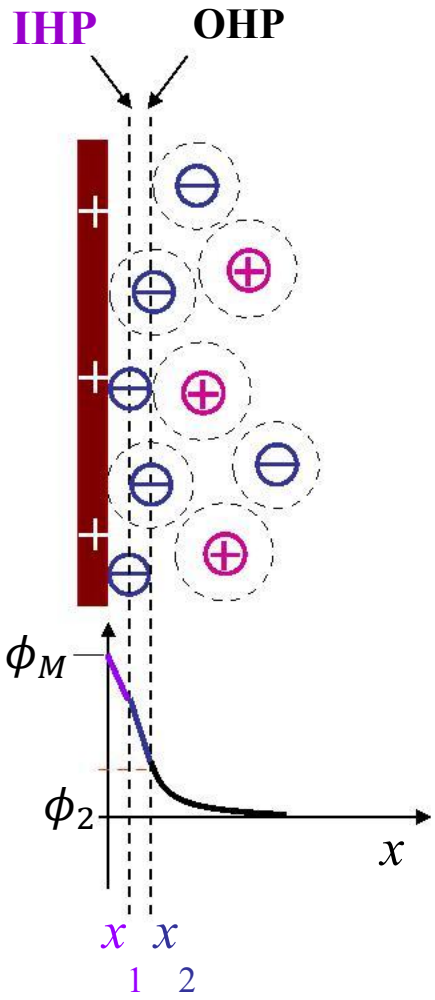


紧密层内无电荷

（ϕ_2与ϕ_M总是同号）

分散层特征参数定量描述

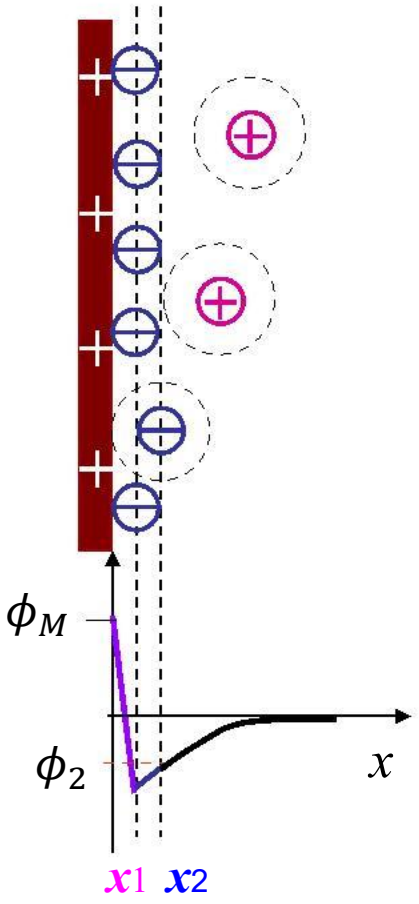
GCS-G双电层模型中的电势分布



阴离子特性吸附

（ϕ_2与ϕ_M同号）

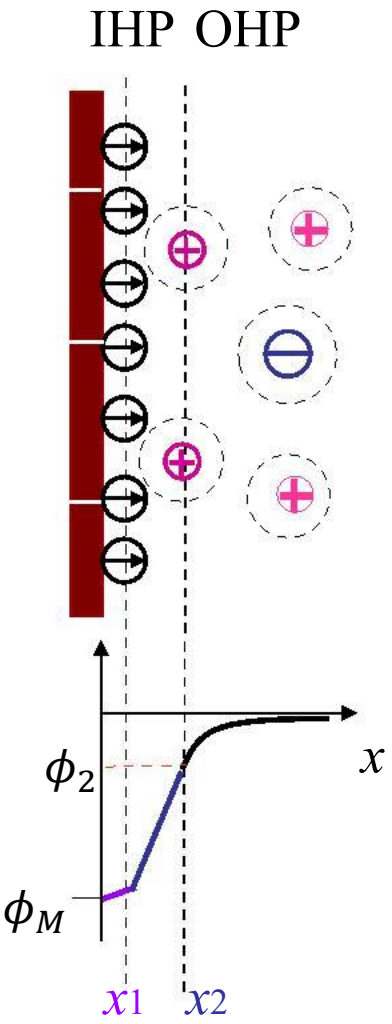
紧密层内有电荷



阴离子特性吸附

(超载吸附, ϕ_2与ϕ_M反号)

紧密层内有电荷

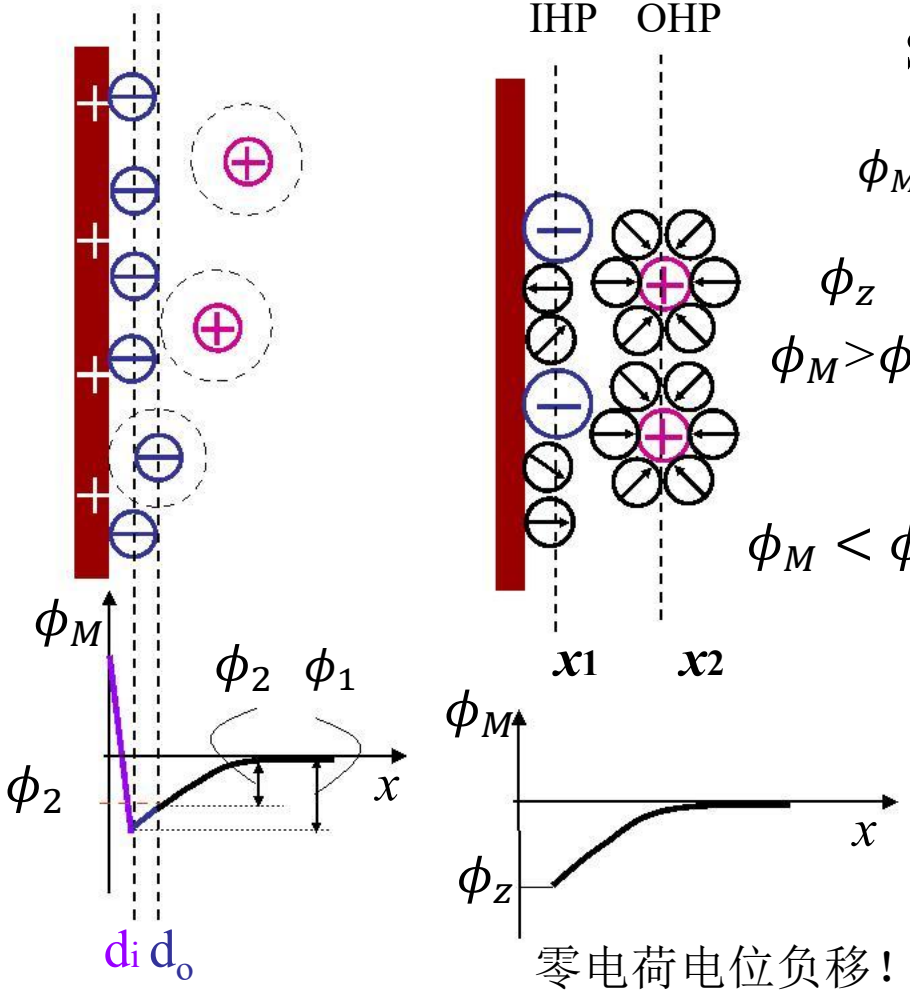


偶极水分子层

紧密层内无电荷

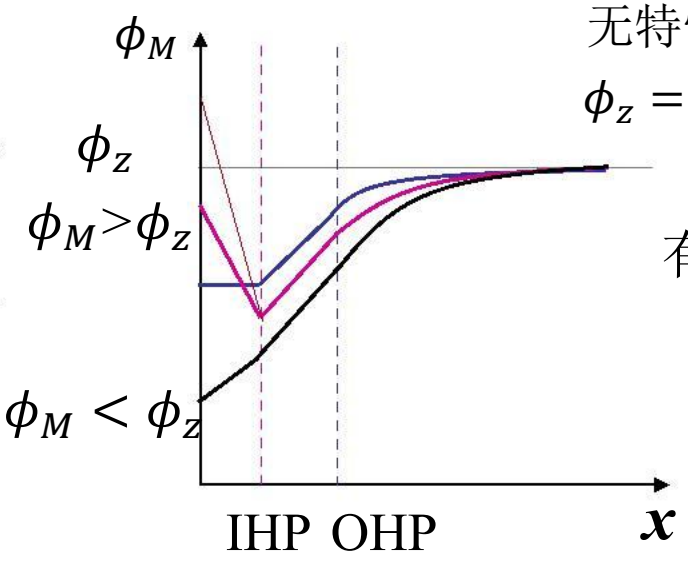
阴离子特性吸附对双电层结构的影响

- 发生超载吸附，形成三电层；



- 零电荷电位顺序负移

$S^{2-} > I^- > Br^- > Cl^- >> OH^- > SO_4^{2-}$ (Hg)



无特性吸附（灰线）

$\phi_z = \phi_1 = \phi_2 = \phi_S$

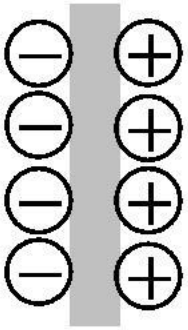
有特性吸附（其它线）

ϕ_2与ϕ_1可能同号，也可能反号

IHP处电势斜率不连续!

金属－电解液界面双电层

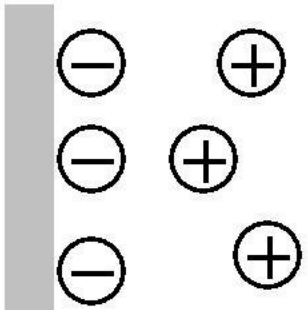
双电层



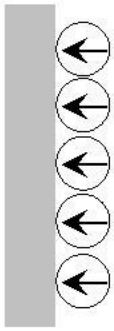
荷电界面

双电层 (electric double layer)
由界面区中金属一侧的剩余电荷和溶液一侧的剩余电荷共同构成, 满足电中性条件, 是电极-溶液界面结构的核心。

吸附双电层

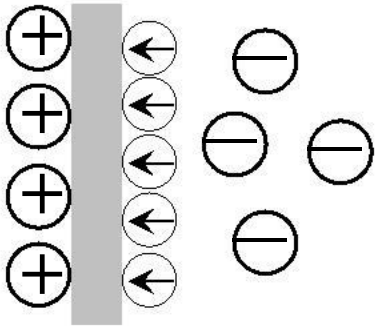


偶极子层



电极表面不带电

偶极子层对DL影响



?

混合型

2.6 零电荷电势 (PZC, E_{PZC}, E_z)

定义: 电极表面自由电荷密度$q_M=0$时的电位, 其数值与电极材料和电解质溶液的组成及双电层结构有关, 是反映界面电性质的重要参量。

- 零总表面电荷电势 (E_{PZTC}, potential of zero total surface charge, **PZTC**)

定义: 存在吸附物种时，需考虑其向表面转移的电荷, 该转移电荷与金属表面自由电荷q_M之和为零时的电位值, 称为零总表面电荷电势.

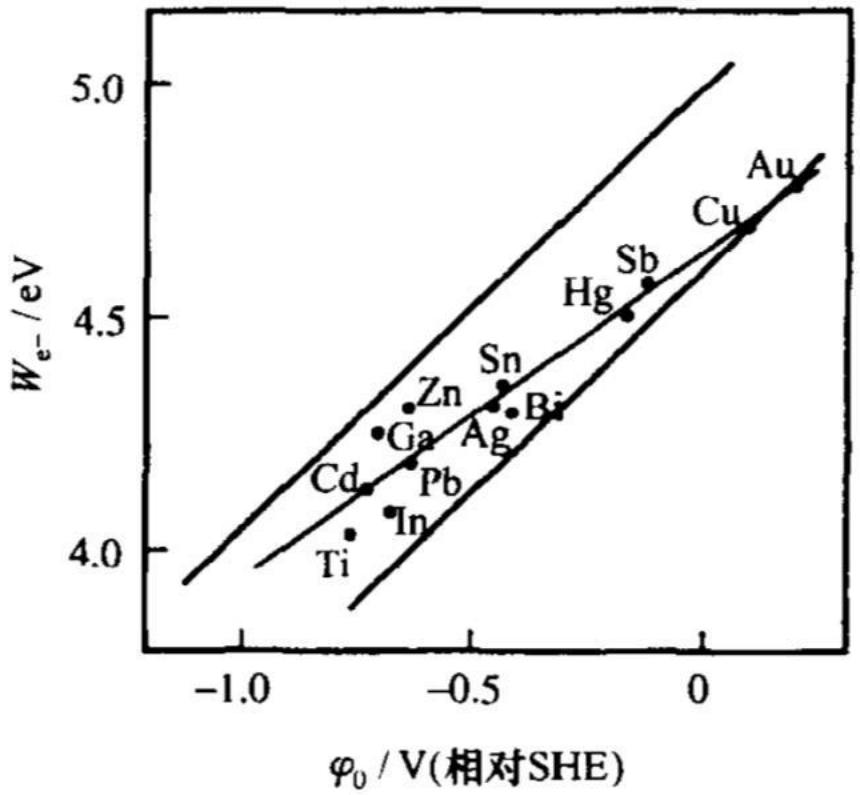
$$E_{pztc} > E_{pzc}$$

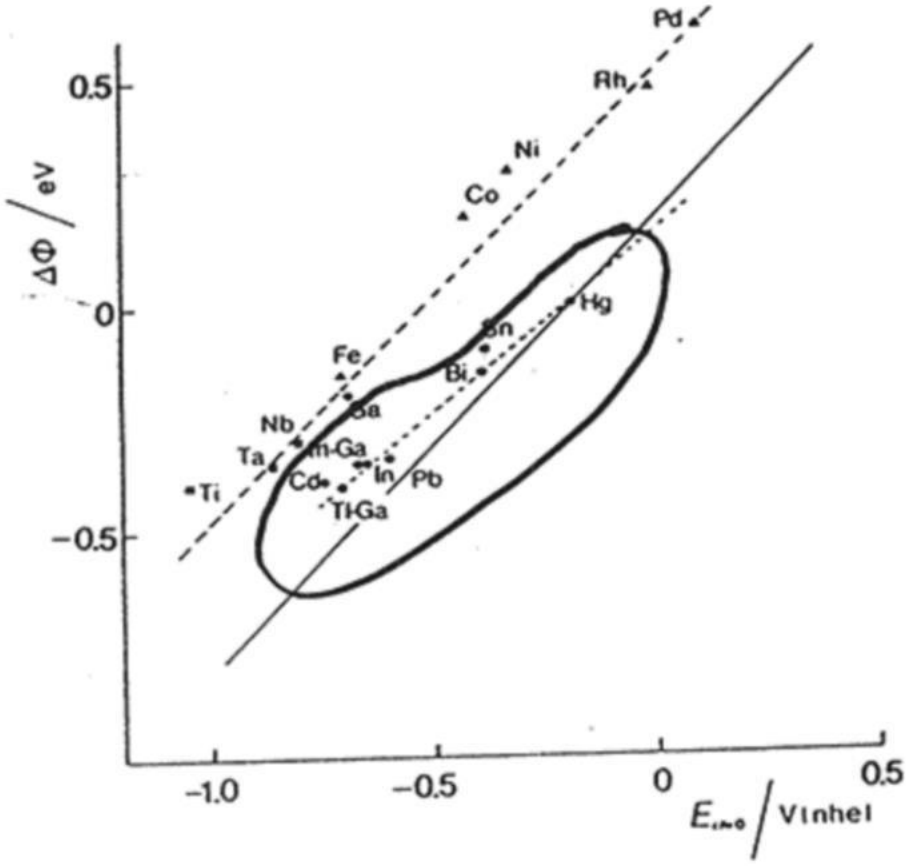
表 2.2　在不含能在电极上有特性吸附的阴离子的稀溶液中测得的 φ_0 值

电极材料	溶液组成	φ_0/V(相对于 SHE)
Hg	0.01mol·L^{-1} NaF	−0.19
Pb	0.01mol·L^{-1} NaF	−0.60
Tl	0.001mol·L^{-1} NaF	−0.71
Cd	0.001mol·L^{-1} NaF	−0.75
Cu	0.01～0.001mol·L^{-1} NaF	+0.09
Ga(液)	0.008mol·L^{-1} $HClO_4$	−0.69
Sb	0.002mol·L^{-1} NaF	−0.15
Sn	0.00125～0.005mol·L^{-1} Na_2SO_4	−0.38
In	0.01mol·L^{-1} NaF	−0.65
Bi(111 面)	0.01mol·L^{-1} KF	−0.42
(多晶)	0.0005mol·L^{-1} H_2SO_4	−0.39
Ag(111 面)	0.001mol·L^{-1} KF	−0.46
(100 面)	0.005mol·L^{-1} NaF	−0.61
(110 面)	0.005mol·L^{-1} NaF	−0.77
(多晶)	0.0005mol·L^{-1} Na_2SO_4	−0.7
Au(110 面)	0.005mol·L^{-1} NaF	+0.19
(111 面)	0.005mol·L^{-1} NaF	+0.50
(100 面)	0.005mol·L^{-1} NaF	+0.38
(多晶)	0.005mol·L^{-1} NaF	+0.25

2.6 零电荷电势 (PZC, E_{PZC}, E_z)

$$W_{e^-}(\text{eV}) = \varphi_0(\text{V，相对 SHE}) + \text{常数}$$

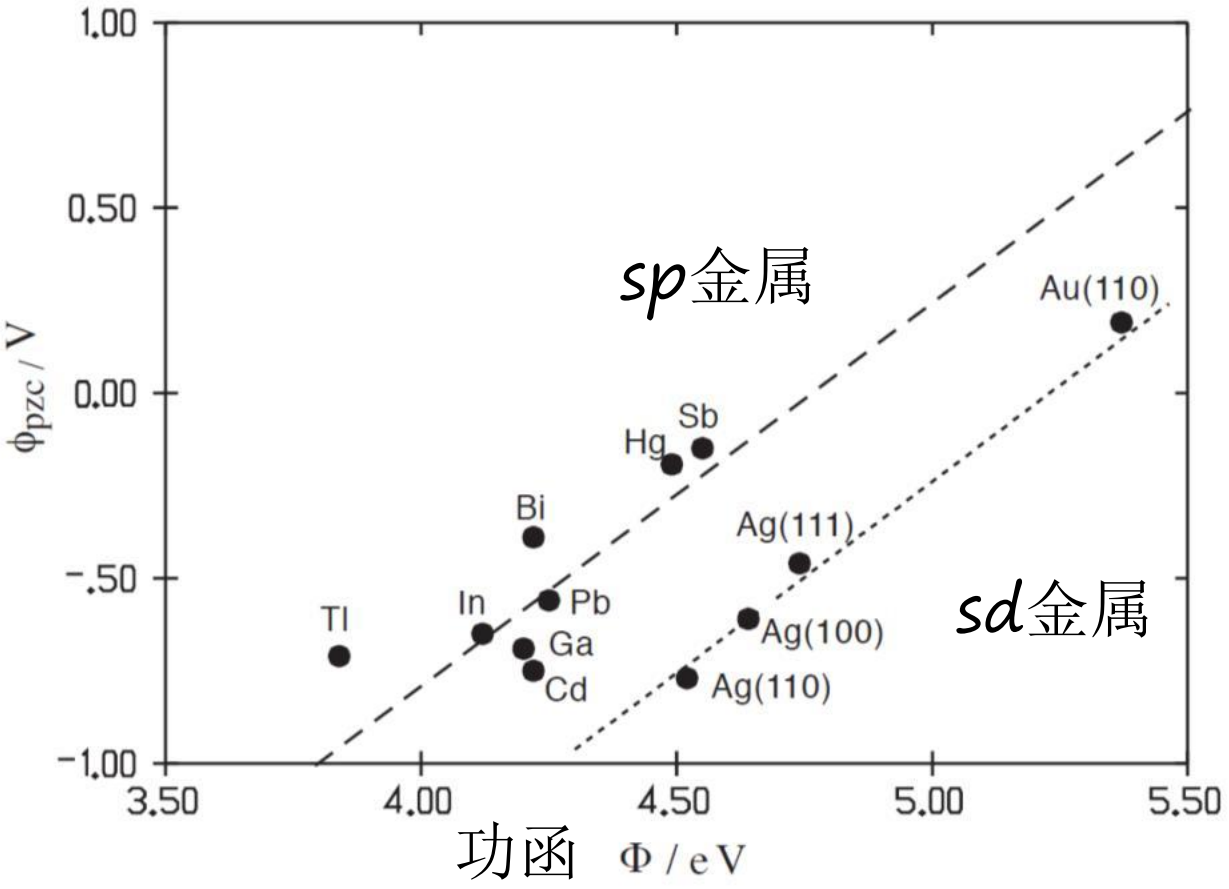




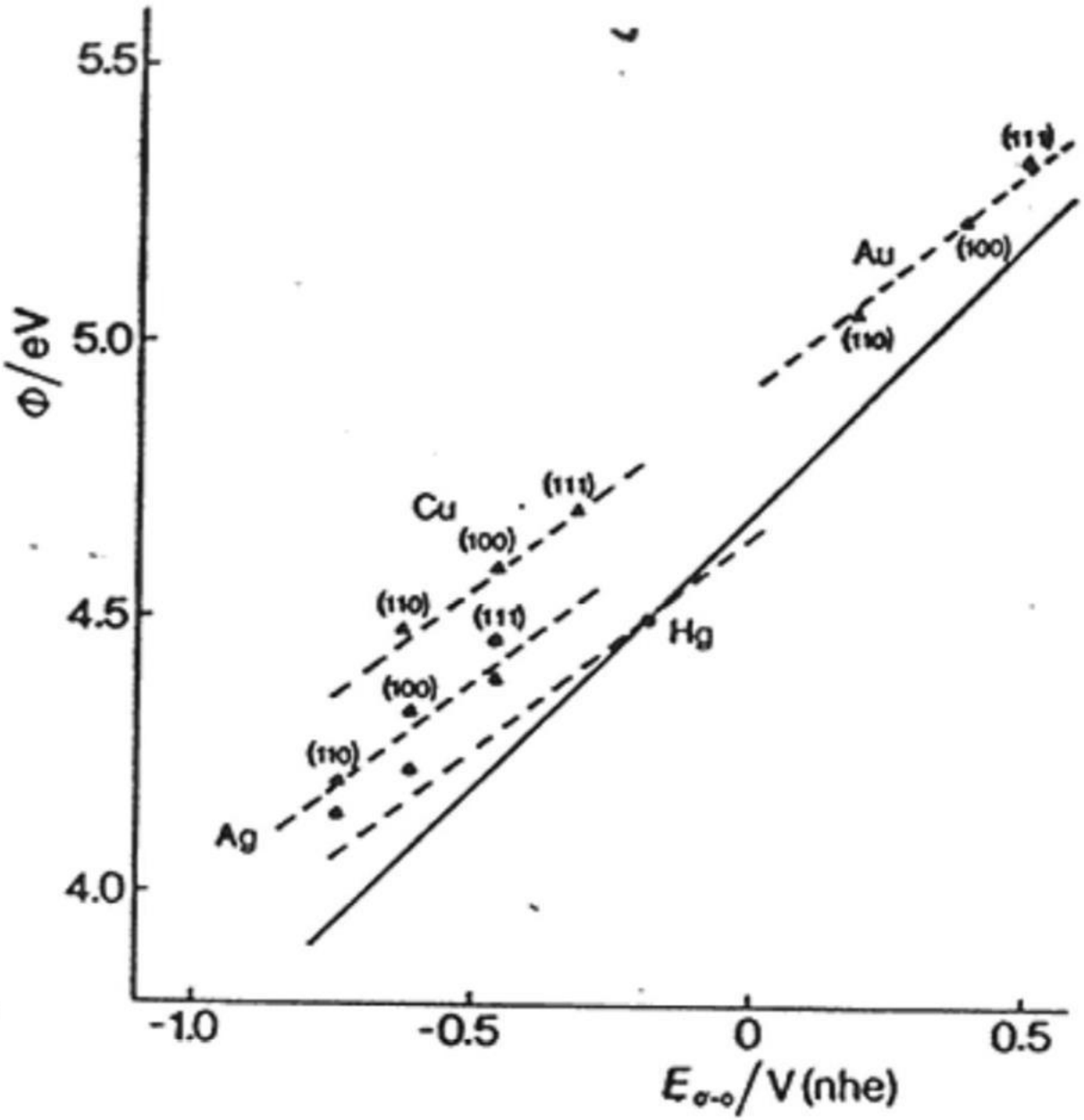
零电荷电位参照到*SHE*

2.6 零电荷电势 (PZC, E_{PZC}, E_z)

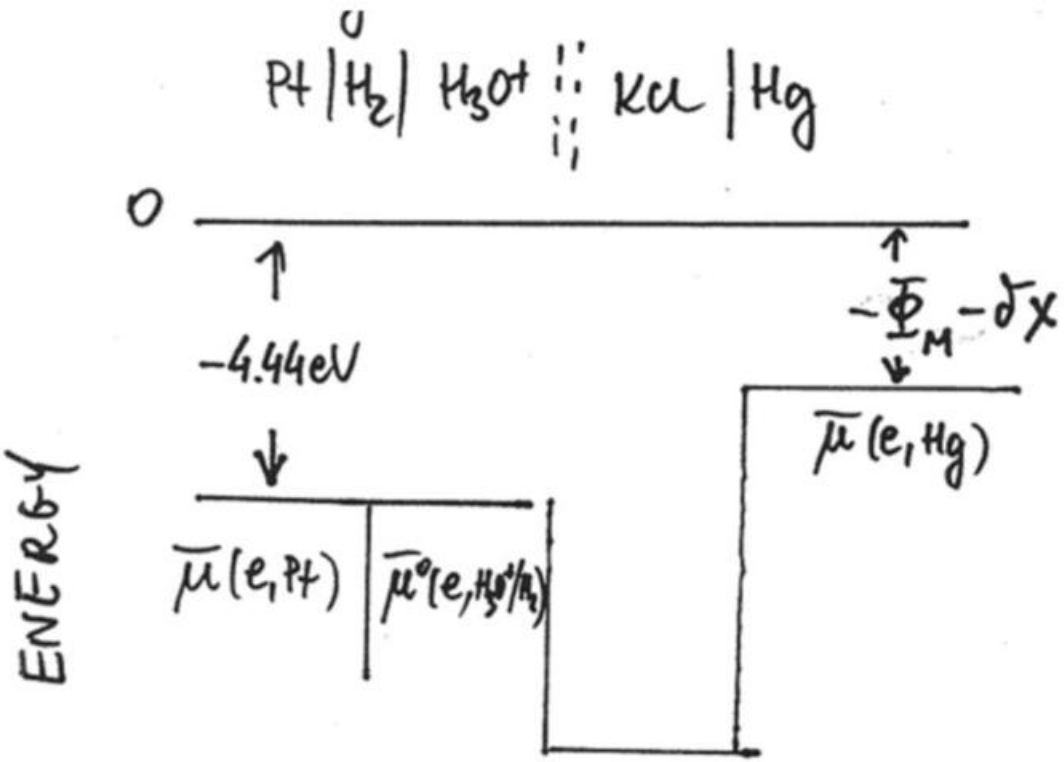
零电荷电位参照到*SHE*



1. W. Schmickler, Interfacial Electrochemistry, Springer-Verlag Berlin Heidelberg 2010.
2. S. Trasatti, Advances in Electrochemistry and Electrochemical Engineering, Vol. 10, edited by H. Gerischer and C.W. Tobias. Wiley Interscience, New York, NY, 1977.


5.5
5.0
4.5
4.0
Φ/eV
-1.0
-0.5
0
0.5
$E_{\sigma-o}$/V(nhe)
Au
(111)
(100)
(110)
Cu
(111)
(100)
(110)
Hg
(111)
(100)
(110)
Ag

2.6 零电荷电势 (PZC, E_{PZC}, E_z)



$$-eE = \bar{\mu}(e,Hg) - \bar{\mu}(e,Pt) - 4.44eV$$

$$\bar{\mu}(e,Hg) = \mu(e,Hg) + e\chi_S^M$$

$$\mu(e,Hg) = \mu(e,Hg) + e\chi_{e+}^M - \chi_S^M$$

$$= -\Phi_M - \delta\chi$$

物理 + 化学

$$-eE = -\Phi_M - e\delta\chi - 4.44eV$$

$$E_{PZC} = \frac{\Phi_M + e\delta\chi}{e} - 4.44V$$

➢ 零电荷电势的含义

- 反映电极表面剩余电荷密度为零时的一种特殊状态, 视电极材料及晶面和电解质溶液的性质及浓度而具有不同的数值.

- 一些界面物理化学性质在**PZC**处出现极值, 如表面张力和微分电容等.

- 零电荷电势 $\mathbf{E_{pzc}}$ 与零电荷相间电势差 $\Delta\phi_{pzc}$ 在数值上相差一常数, 故 $\mathbf{E_{pzc}} \neq \Delta\phi_{pzc}$

$\Delta\phi_{pzc}$

- 零电荷相间电势差 j_{pzc} 不是界面电势差的零点, 可因偶极溶剂分子的取向排列或离子特性吸附造成表面电势差.

- 零荷电势不同于平衡电势, 电极反应可在**PZC**时进行.

- **Esin-Markov** 效应 发生阴离子特性吸附时,零电荷电势负移, 同种电解质中零电荷电势随浓度增加而负移.

ELSEVIER

Journal of Electroanalytical Chemistry 414 (1996) 53–59

JOURNAL OF ELECTROANALYTICAL CHEMISTRY

A new method for rapid determination of the potential of zero charge for gold|solution interfaces

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Received 25 January 1996; revised 7 March 1996

Abstract

A new method, based on the response of a thickness shear mode sensor to electrocapillary phenomena, for determination of the potential of zero charge (pzc) at solid metal|solution interfaces is described. The pzc values of gold in the solutions of NaF, Na_2SO_4, and $HClO_4$ are obtained. The order of the pzc values is as follows: $F^- > ClO_4^- > SO_4^{2-}$. This is in good agreement with previous work.

Table 1
Pzc values for gold, determined by TSM sensor in various electrolyte solutions

Solution		Pzc/V	
		(vs. SCE)	(vs. SHE)
$HClO_4$	1 mM	−0.034	+0.207
	5 mM	−0.064	+0.177
	10 mM	−0.086	+0.155
NaF	1 mM	+0.039	+0.280
	5 mM	+0.032	+0.273
	0.1 M	+0.037	+0.278
Na_2SO_4	0.5 mM	−0.134	+0.107
	0.1 M	−0.151	+0.090

Table 2
Comparison of values of the pzc for gold determined by various methods

Solution	Crystal type [a]	Pzc/V (vs. SHE)	Method	Reference
$HClO_4$				
0.001 M	P	0.207	TSM	This work
0.01 M	P	0.155	TSM	This work
0.001 M	P	0.201	Capacitance method	[28]
0.001 M	P	0.340	Charging current of freshly immersed metal surface	[36]
0.075 M	P	0.000	Bending of thin metal electrode	[9]
0.001 M	P	0.170	Capacitance method	[11]
0.01 M	P	0.150	Friction method	[11]
0.1 M	P	0.100	Friction method	[11]
0.1 M	P	0.140	Gravimetric measurement of solution meniscus	[1]
0.01 M	S[100]	0.190	Capacitance method	[37]
0.01 M	S[100]	0.240	Capacitance method	[37]
1.00 M	P	−0.160	Extension of length of metal foil	[38]
NaF				
0.005 M	P	0.273	TSM	This work
0.005 M	S[110]	0.190	Capacitance method	[39]
0.005 M	S[111]	0.500	Capacitance method	[39]
0.005 M	S[110]	0.380	Capacitance method	[39]
0.005 M	P	0.250	Capacitance method	[39]
Na_2SO_4				
0.0005 M	P	0.107	TSM	This work
0.1 M	P	0.090	TSM	This work
0.1 M	P	0.140	Open-circuit potential of fresh metal surface	[40]
0.01 M	P	0.230	Capacitance method	[41]
0.005 M	P	0.090	Capacitance and open-circuit potential of fresh metal surface	[42]
0.01 M	S[100]	0.020	Capacitance method	[43]
0.01 M	S[110]	0.100	Capacitance method	[43]
0.01 M	S[111]	0.450	Capacitance method	[43]

[a] P, polycrystalline; S, single crystal.

JinHua Chen, LiHua Nie, ShouZhuo Yao, J. Electroanal. Chem., 1996, 414 (1996) ,53-59.

J. Electroanal. Chem., 294 (1990) 239–251
Elsevier Sequoia S.A., Lausanne

The temperature coefficient of the potential of zero charge of the gold single-crystal electrode / aqueous solution interface

Possible relevance to gold–water interactions

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(Received 14 March 1990; in revised form 4 June 1990)

ABSTRACT

The minimum in the differential capacity of (111), (100), (110), (210) and (311) single-crystal gold faces in contact with dilute solutions of $HClO_4$ as a function of the temperature was used to estimate the temperature coefficient of the potential of zero charge of these interfaces. The results are discussed in terms of the interaction of water with the metallic surfaces and the role played by the surface atomic arrangement. The strength of Au (*hkl*)–water interactions are proposed to be in the order Au (111) > Au (311) > Au (100) > Au (110) > Au (210) ≈ Hg.

晶面与温度效应

Au（100）

Au（110）

Au（111）

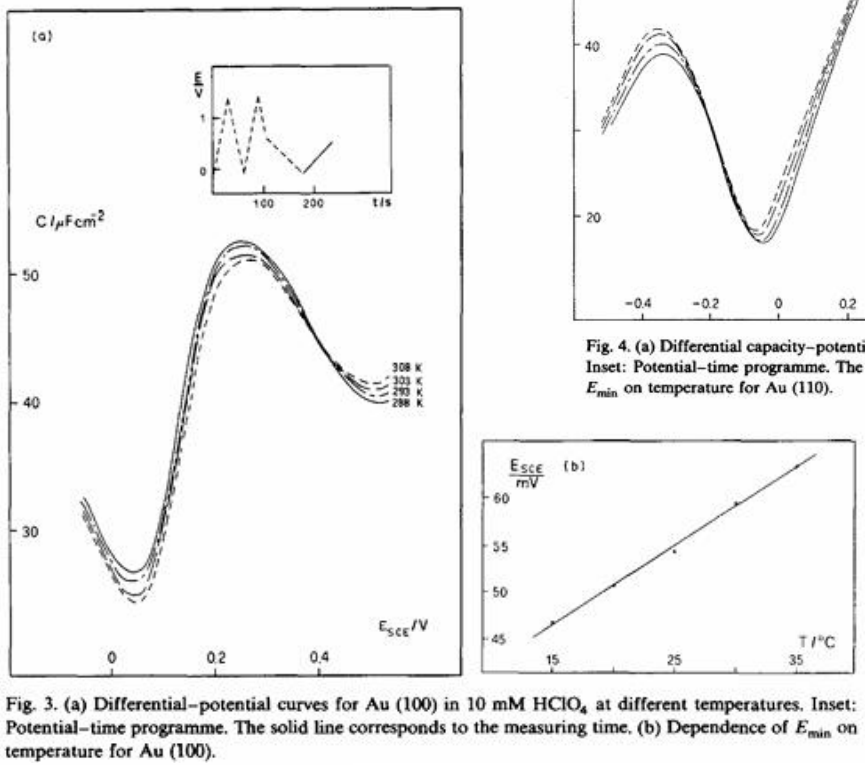


Fig. 3. (a) Differential–potential curves for Au (100) in 10 mM $HClO_4$ at different temperatures. Inset: Potential–time programme. The solid line corresponds to the measuring time. (b) Dependence of E_{min} on temperature for Au (100).

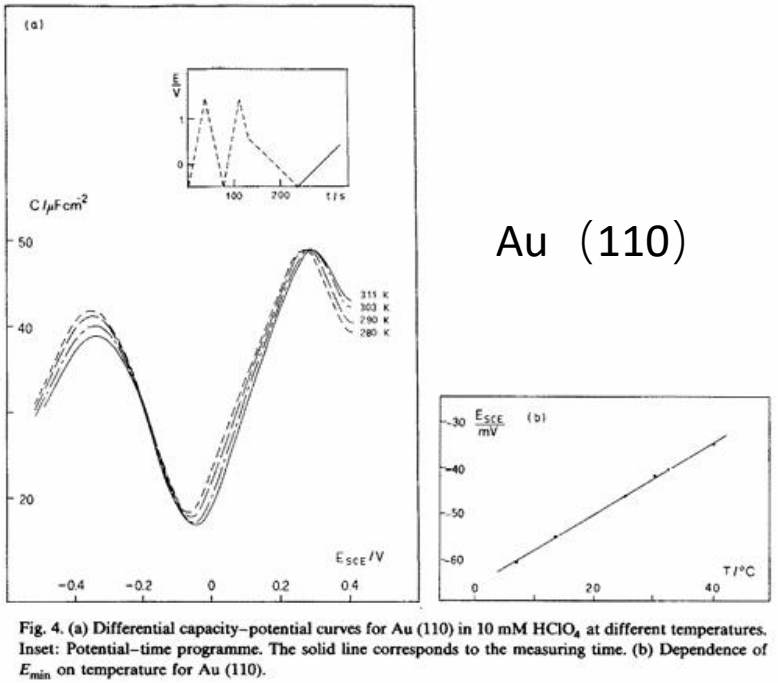


Fig. 4. (a) Differential capacity–potential curves for Au (110) in 10 mM $HClO_4$ at different temperatures. Inset: Potential–time programme. The solid line corresponds to the measuring time. (b) Dependence of E_{min} on temperature for Au (110).

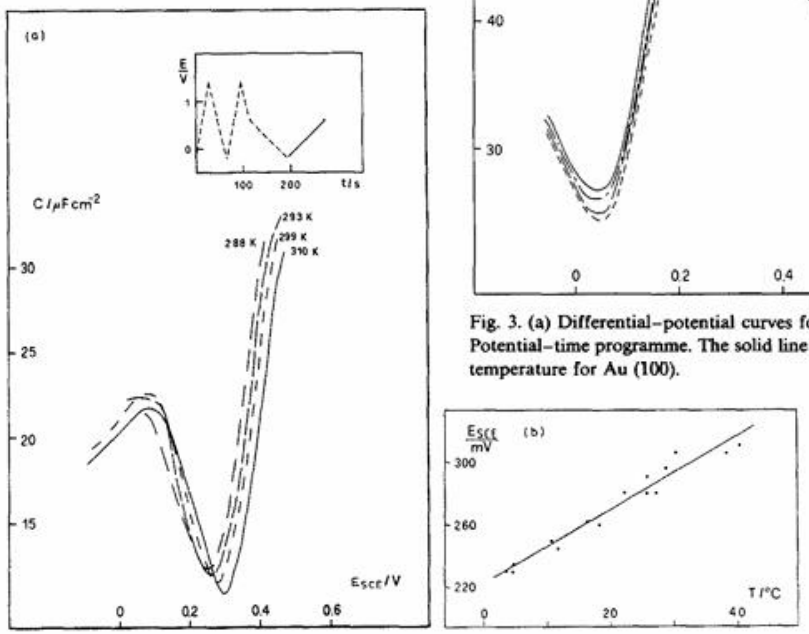


Fig. 2. (a) Differential capacity–potential curves for Au (111) in 1 mM $HClO_4$ at different temperatures. Inset: Potential–time programme. The solid line corresponds to the measuring time. (b) Dependence of E_{min} on temperature for Au (111).

Fernando Silva, M.J. Sottomayor, A. Hamelin, J. Electroanal. Chem., 294 (1990) 239-251.

晶面与温度效应

Au（210）

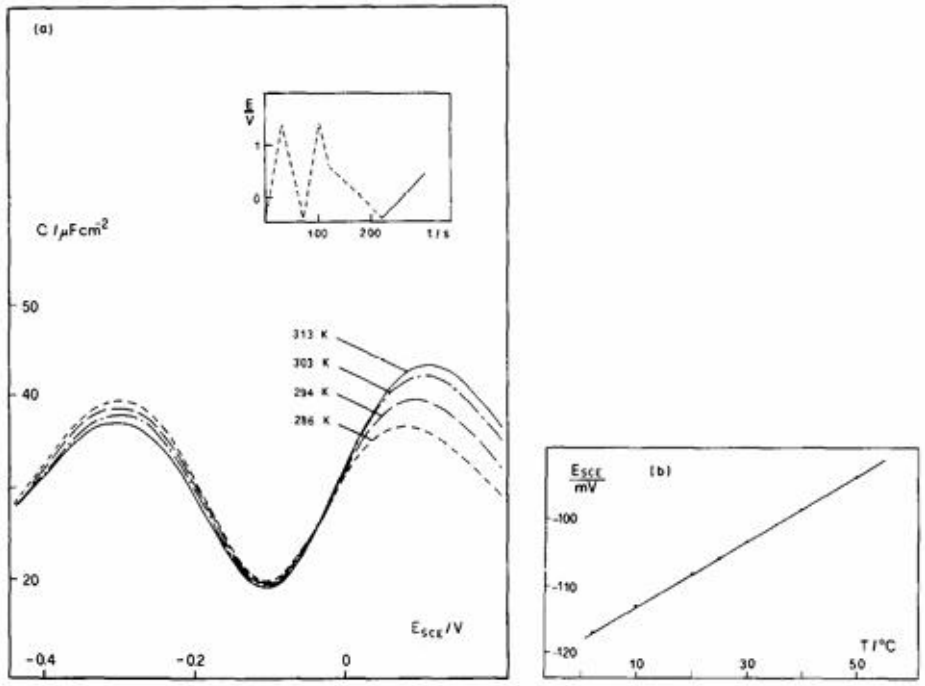


Fig. 5. (a) Differential capacity–potential curves for Au (210) in 10 mM $HClO_4$ at different temperatures. Inset: Potential–time programme. The solid line corresponds to the measuring time. (b) Dependence of E_{min} on temperature for Au (210).

Au（311）

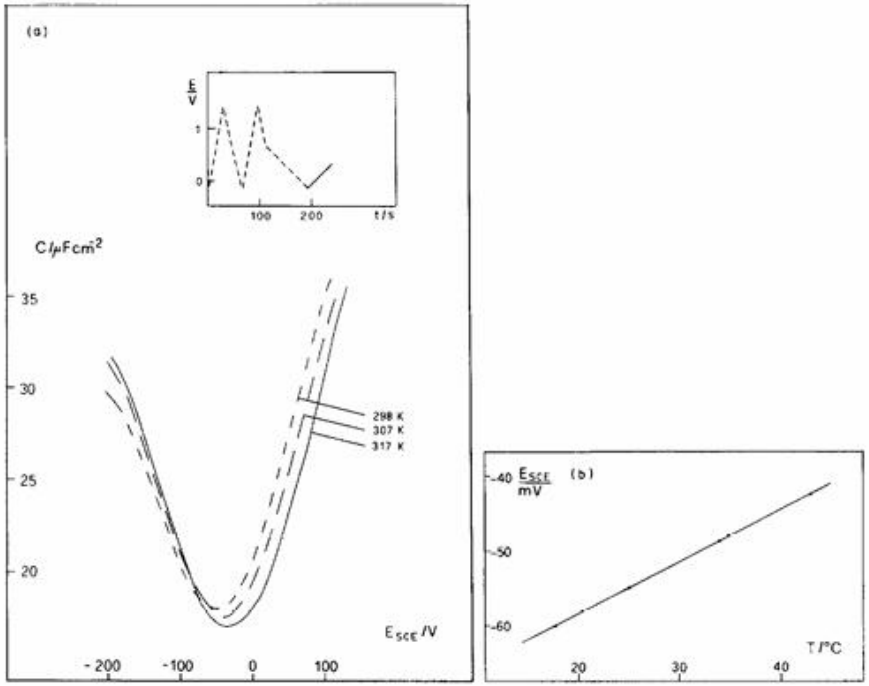


Fig. 6. (a) Differential capacity–potential curves for Au (311) in 5 mM $HClO_4$ at different temperatures. Inset: Potential–time programme. The solid line corresponds to the measuring time. (b) Dependence of E_{min} on temperature for Au (311).

Fernando Silva, M.J. Sottomayor, A. Hamelin, J. Electroanal. Chem., 294 (1990) 239-251.

金表面的晶面与温度效应

TABLE 1

Temperature coefficients of the potential of zero charge and of the differential capacity minimum of gold single-crystal electrode surfaces in contact with dilute $HClO_4$ solutions

	Au (111)	Au (100)	Au (110)	Au (210)	Au (311)	Hg
c_{HClO_4}/mM	3	10	10	10	5	10
(dC_{min}/dT) /μF cm^{-2} K^{-1}	−0.12	−0.08	−0.06	−0.03	−0.06	
(dE_{min}/dT) /mV K^{-1}	2.3	1.0	0.9	0.5	1.3	0.57 [a]

[a] From ref. 10.

Fernando Silva, M.J. Sottomayor, A. Hamelin, J. Electroanal. Chem., 294 (1990) 239-251.

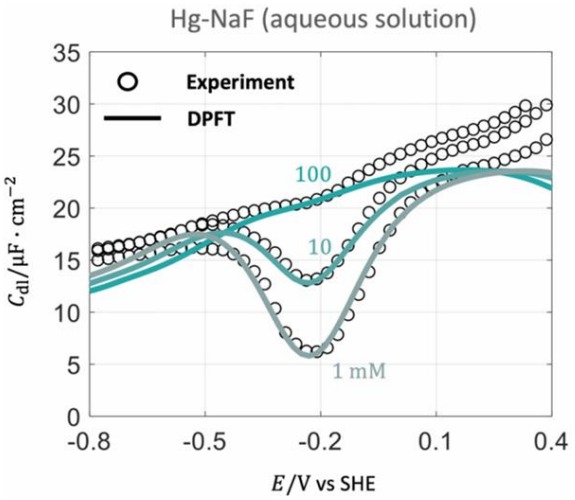
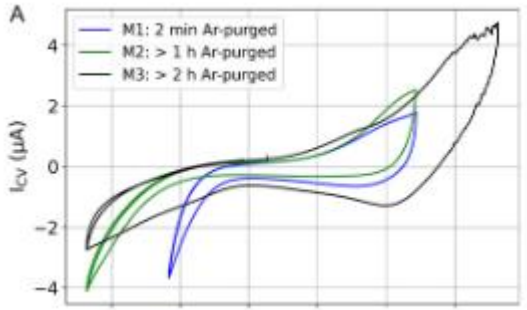
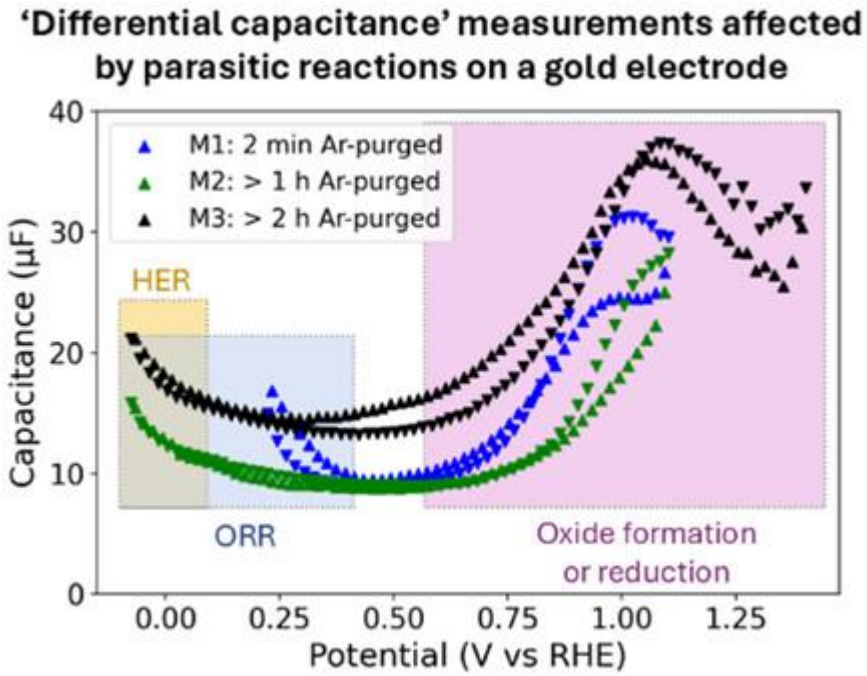


Figure 4. Comparison of C_{dl} between experimental results (circle) and DPFT model results (solid line) in Hg-NaF aqueous solution at varying ion concentrations. Calibrated parameters are $\theta_T = 1.53$, $\epsilon_{op} = 3.70$, $\beta_1 = 1$, $D_{ma(c)} = D_{ms}/6$, and $d_{ma(c)} = 7.56a_0$. Other parameters are given in Supporting Information. Experimental data are obtained from Grahame et al.[33] The electrode potential is transformed to the SHE scale.





Zhang, ZM and Huang, J, Microstructure of Electrical Double Layers at Highly Charged States, JACS AU 2025, 5 (7) , 3453-3467(Jul 28).

Schalenbach, M; Tempel, H and Eichel, RA, Capacitance Measurements for Evaluating Electrochemical Double-Layer Models and Potentials of Zero Charge: A Reassessment, CHEMPHYSCHEM 2025, 26 (14), DOI 10.1002/cphc.202401088(Jul 18)

The surface potential of zero charge (pzc) controled the kinetics and the reaction mechanism of diazonium salts electropolymerization. The kinetics of the surface reaction and the reduction potential depended on the pzc of the surface following the order of gold>platinum>carbon>indium-tin-oxide>fluorine-doped tin-oxide>silicon hydride.



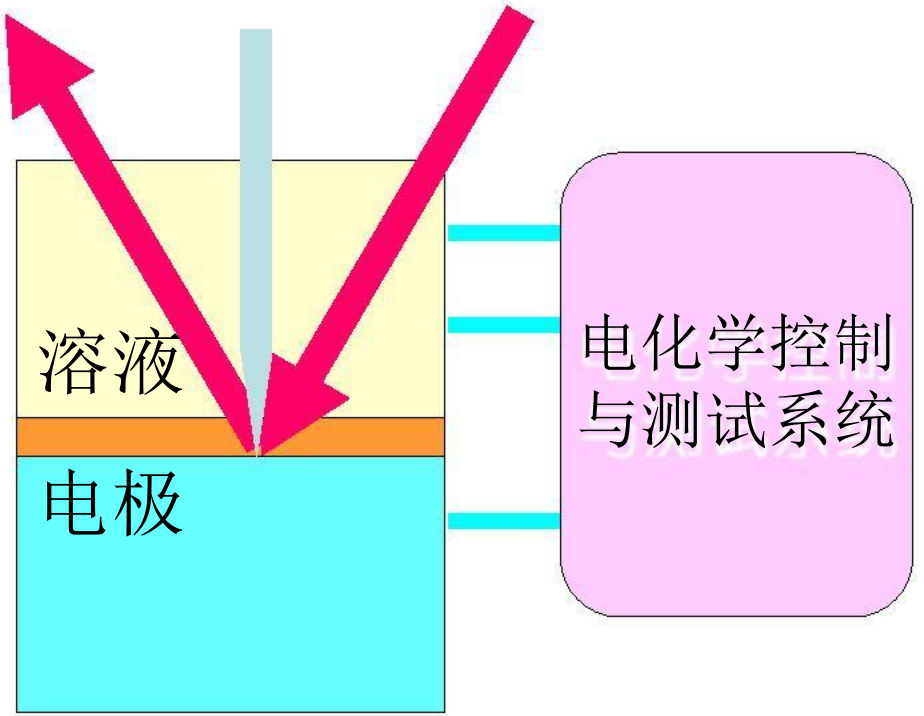
Tiexin Li, Dr. Simone Ciampi, Dr. Nadim Darwish, The Surface Potential of Zero Charge Controls the Kinetics of Diazonium Salts Electropolymerization, 2022, 9(17), E202200255.

谱 学 电 化 学
Spectro-electrochemistry

电化学研究方法

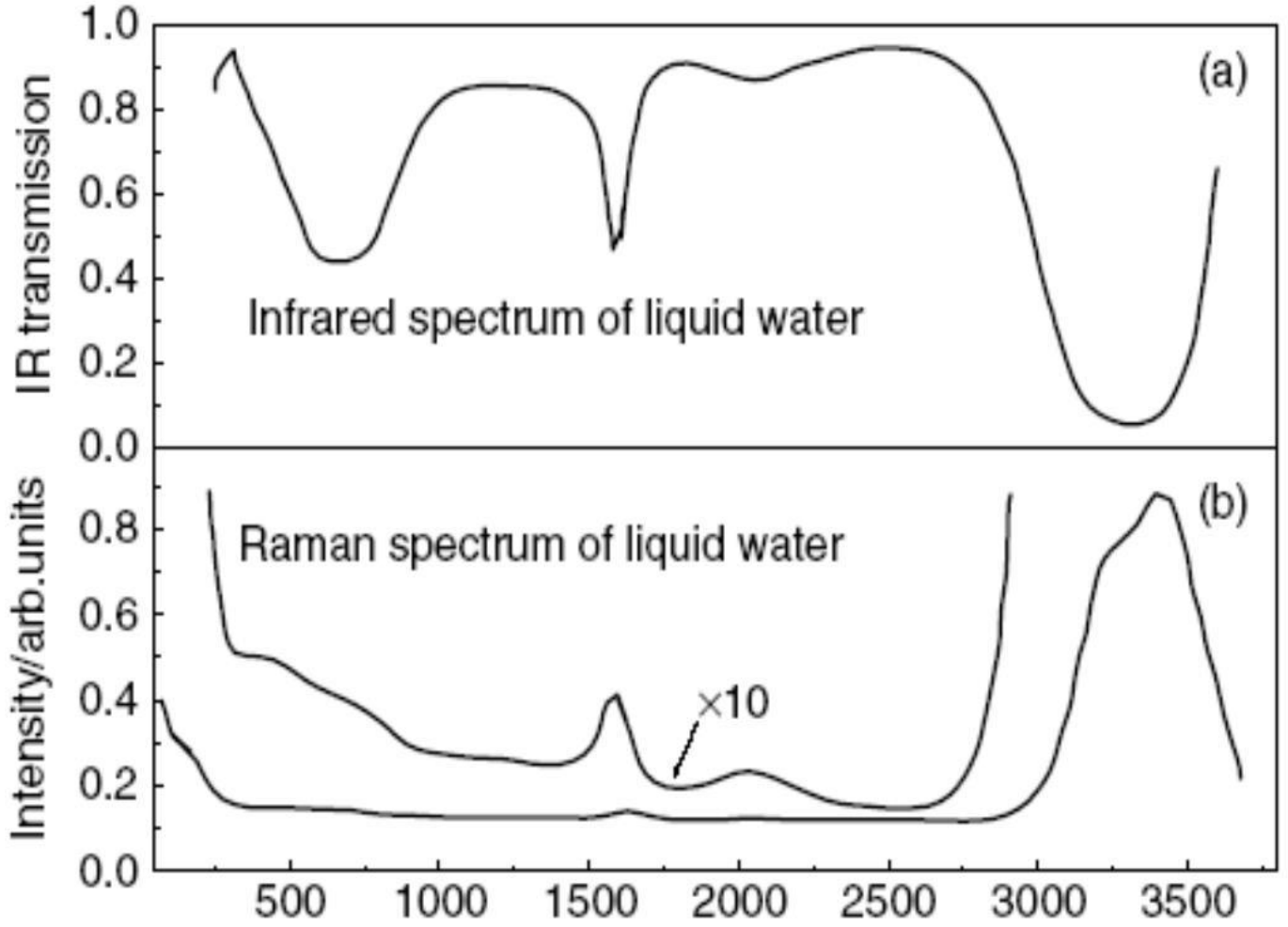
光 谱 方 法1960

扫描探针显微术1980



光谱学技术应用于电化学界面

-------从分子水平研究电化学界面



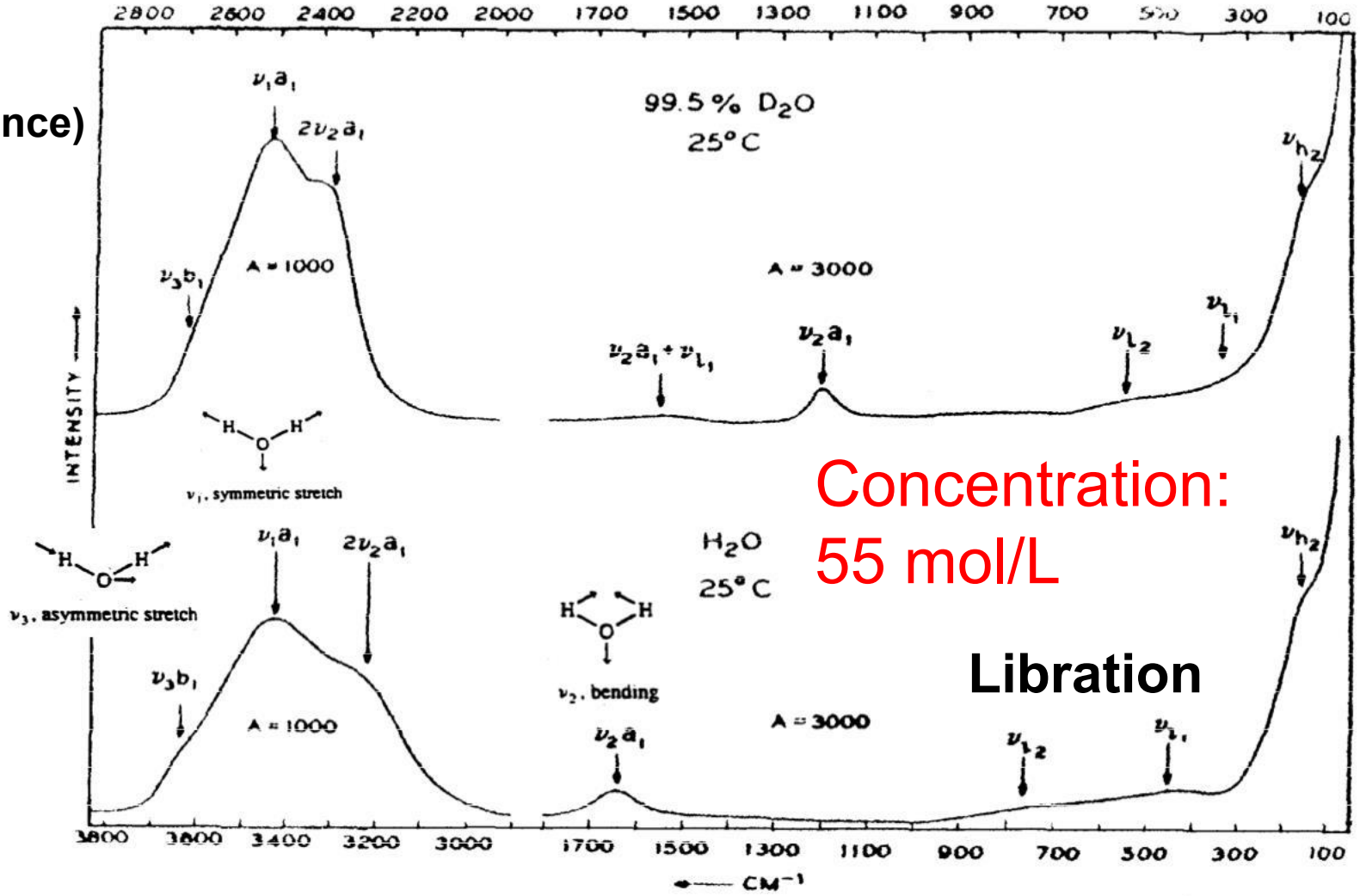
Electronic effects in SERS by liquid water

Yan-Xia Chen† and Andreas Otto*

J. Raman Spectrosco., 2005, 36, 736-747

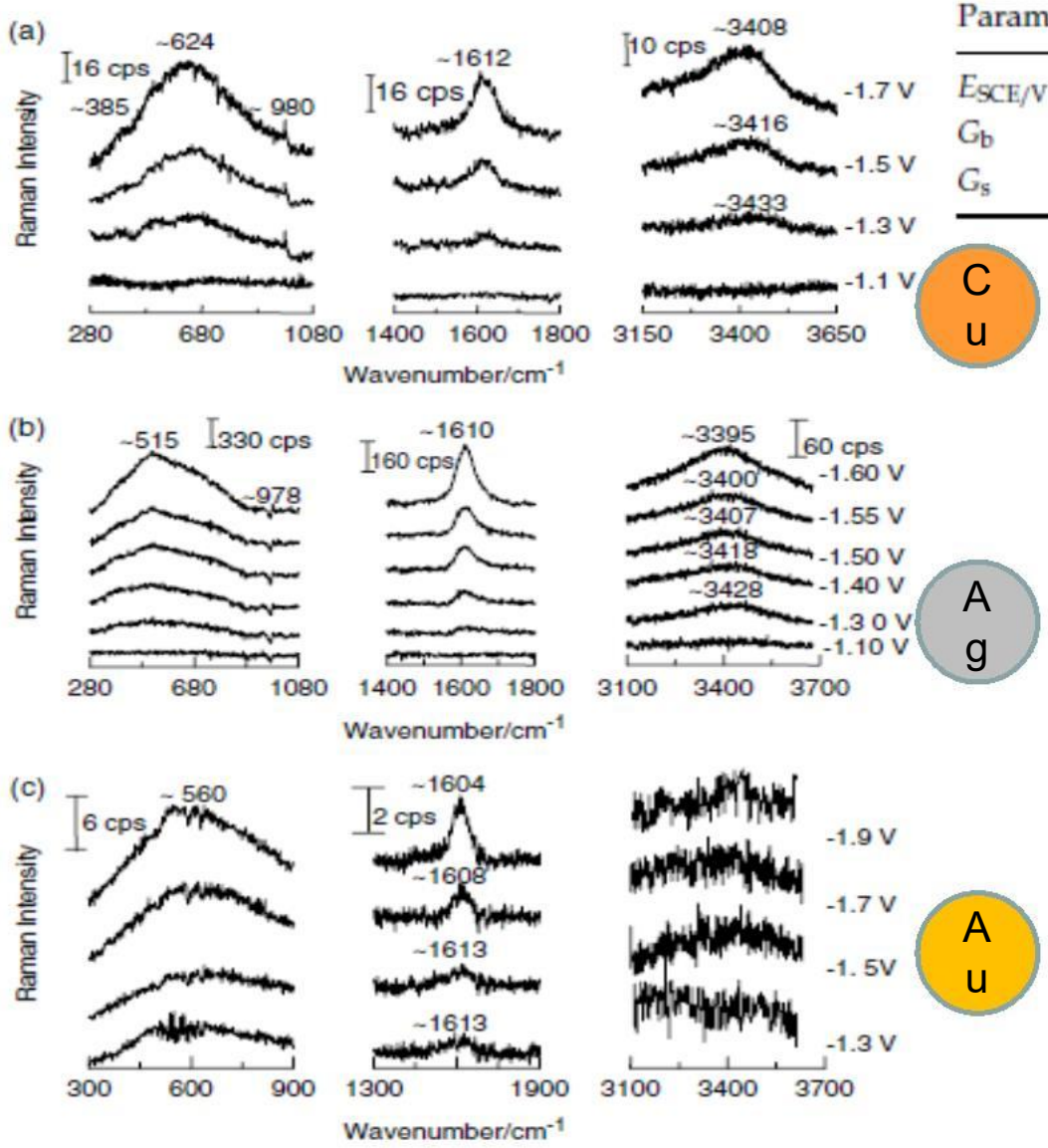
Raman spectra of water: H_2O and D_2O

(Non-resonance)



是否有？；有什么？；有多少？

界面水的SERS表征



Parameter	Ag		Cu		Au
$E_{SCE/V}$	−1.5	−1.6	−1.5	−1.7	−1.7
G_b	4.2×10^6	1.5×10^7	3.6×10^5	1.6×10^6	8.4×10^4
G_s	7.3×10^4	1.5×10^5	2.0×10^4	6.1×10^4	182

1.

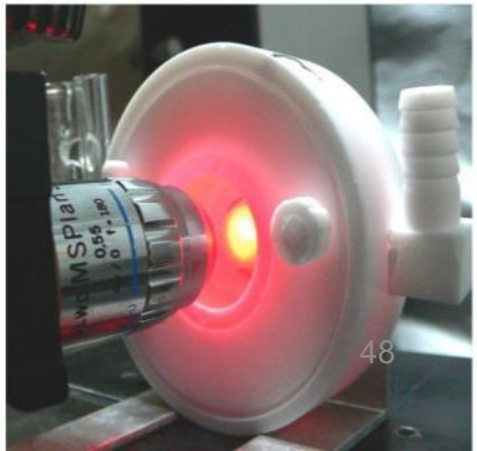
弯曲、伸缩振动显著增强，

高2个数量级

弯曲振动的增强因子比伸缩振动

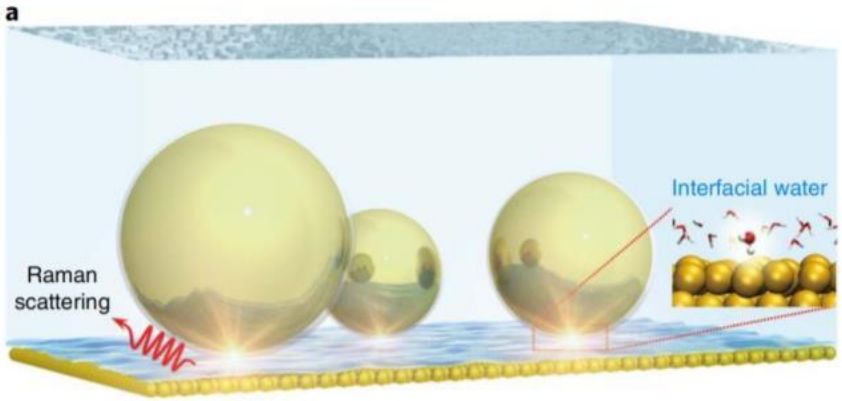
2. 负电势区域，析氢电位

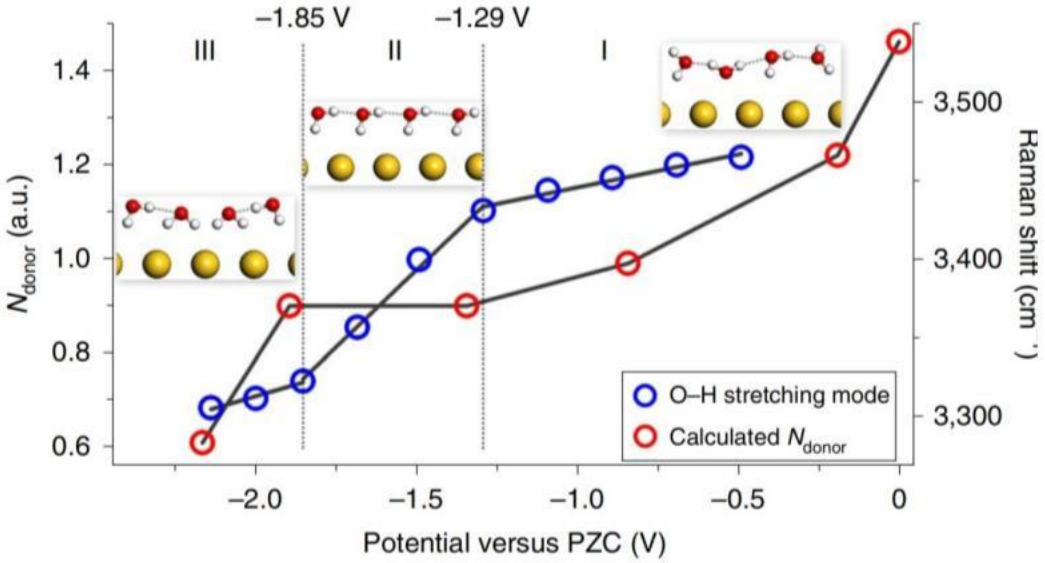
3.粗燥电极上，无法获得过渡金属上水的SERS信号

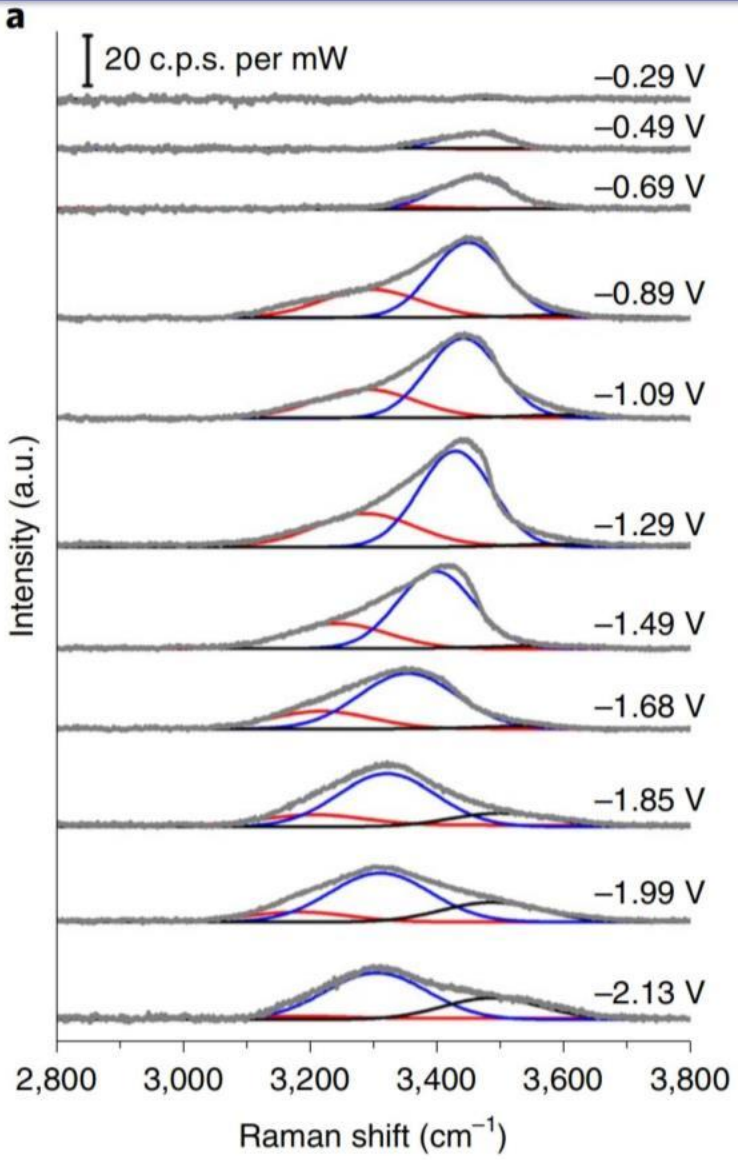


Yan-Xia Chen et.al., *J. Raman. Spectrosc.*, 2005, 36,

在光子光滑表面原位探测电场界面水的吸附结构







C. Y. Li, et al, Nature Materials, 2019,18,

（5）表面活性粒子在“电极/溶液”界面上的吸附

表面活性粒子 (包括无机物或有机物、离子或分子型等):

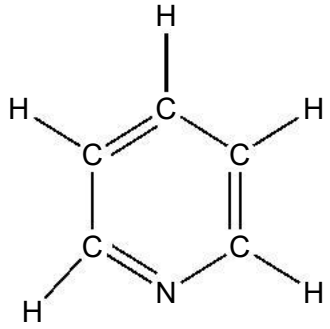
- 不参加反应的（居外）物种
  - 可改变电极表面状态和双电层中的电势分布，
  - 影响反应粒子的表面浓度
  - 反应活化能
- 反应粒子或反应产物（包括中间粒子）
  - 影响自身的动力学过程

可发生吸附的有机分子类型：

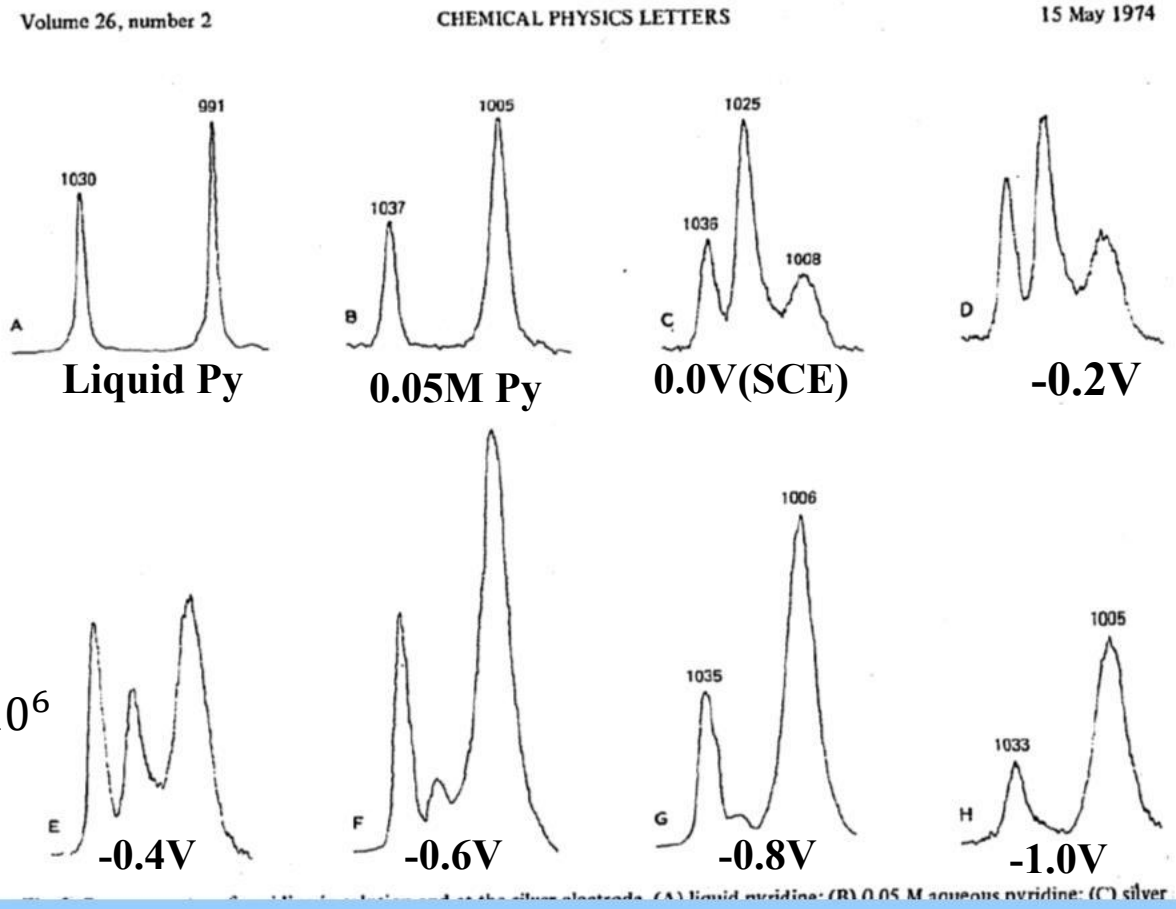
- “阴离子型”活性物质 – 有机磺酸盐和硫酸盐；
- “阳离子型”活性物质- 各种季铵盐；
- “非离子型”- 环氧乙烷与高级醇的缩聚物等

查全性书 2.7（p56）；
吴辉煌教材 p18-19, 1.3.2

吡啶吸附于粗糙银电极的SERS光谱



$$EF = \frac{I_{SERS}/N_{SERS}}{I_{Bulk}/N_{Bulk}} \approx 10^6$$

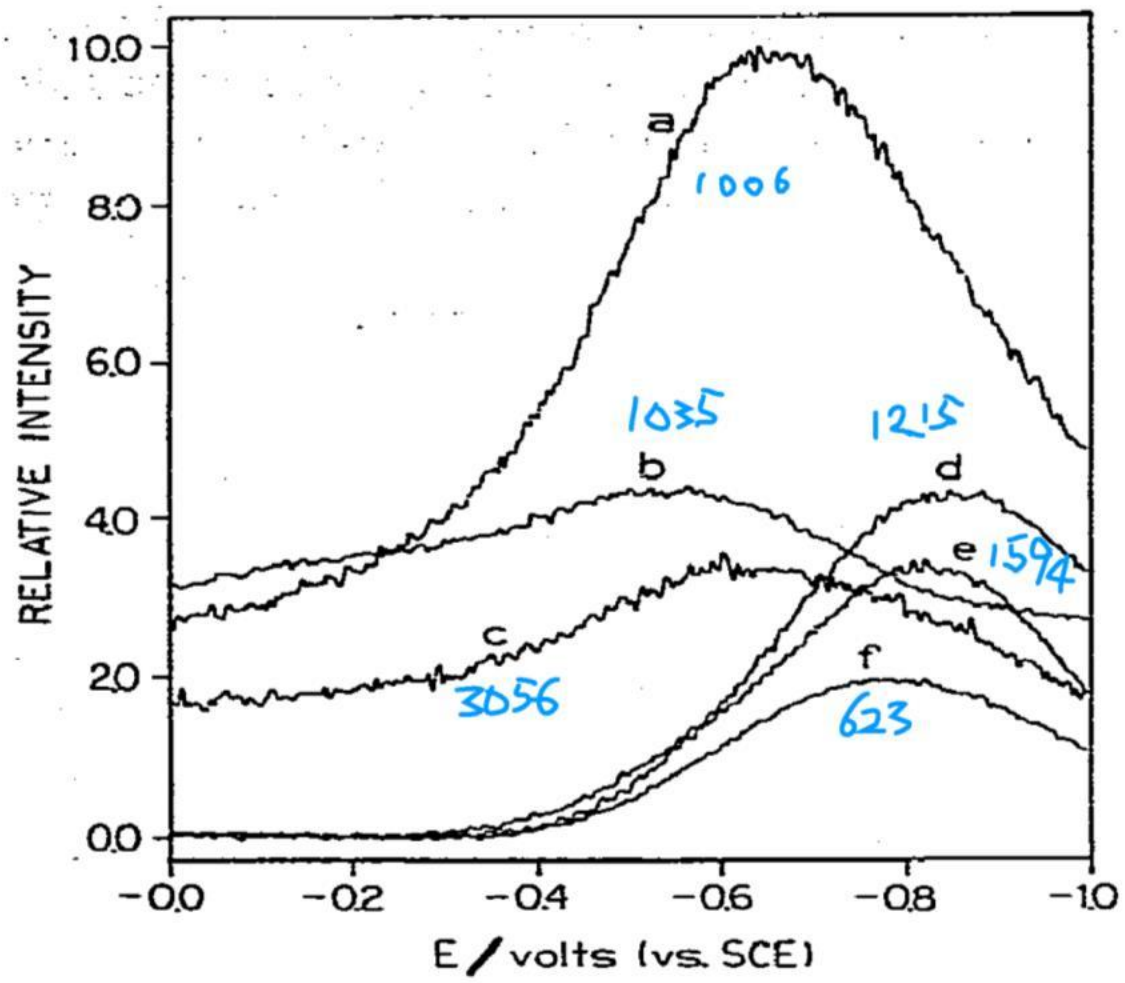


1974 M Fleischmann, PJ Hendra, AJ McQuillan, Pyridine/roughed silver electrode surfaces

1977 DL Jeanmaire, RP van Duyne, surface-enhanced Raman scattering effect

1977 MR Albrecht, JA Creighton, et al. Anomalously intense Raman spectra of pyridine o a silver electrode

吡啶吸附于粗糙银电极的SERS光谱

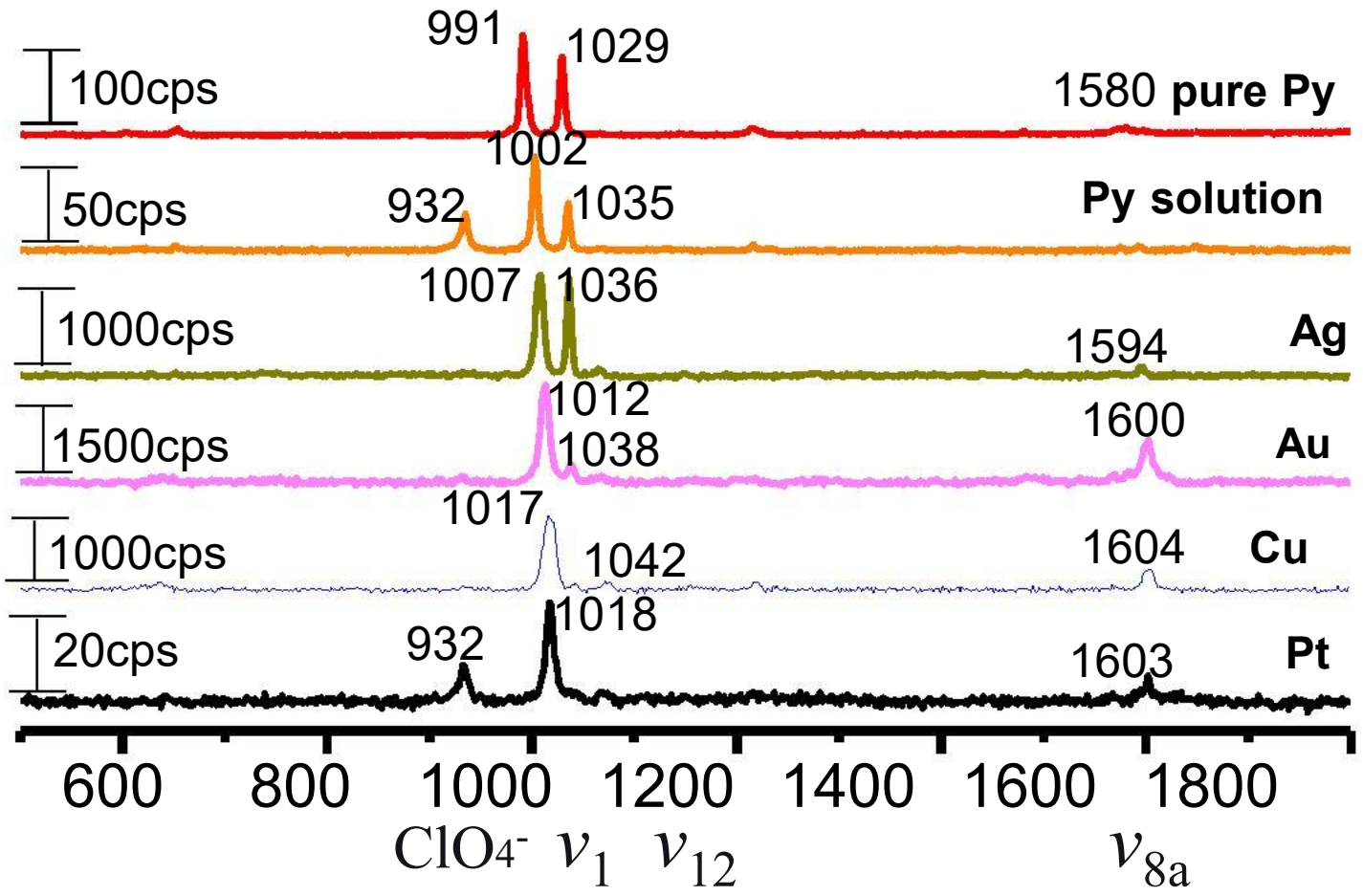


吡啶/银电极：514.5nm

J. Electroanal. Chem. 1977,84,1

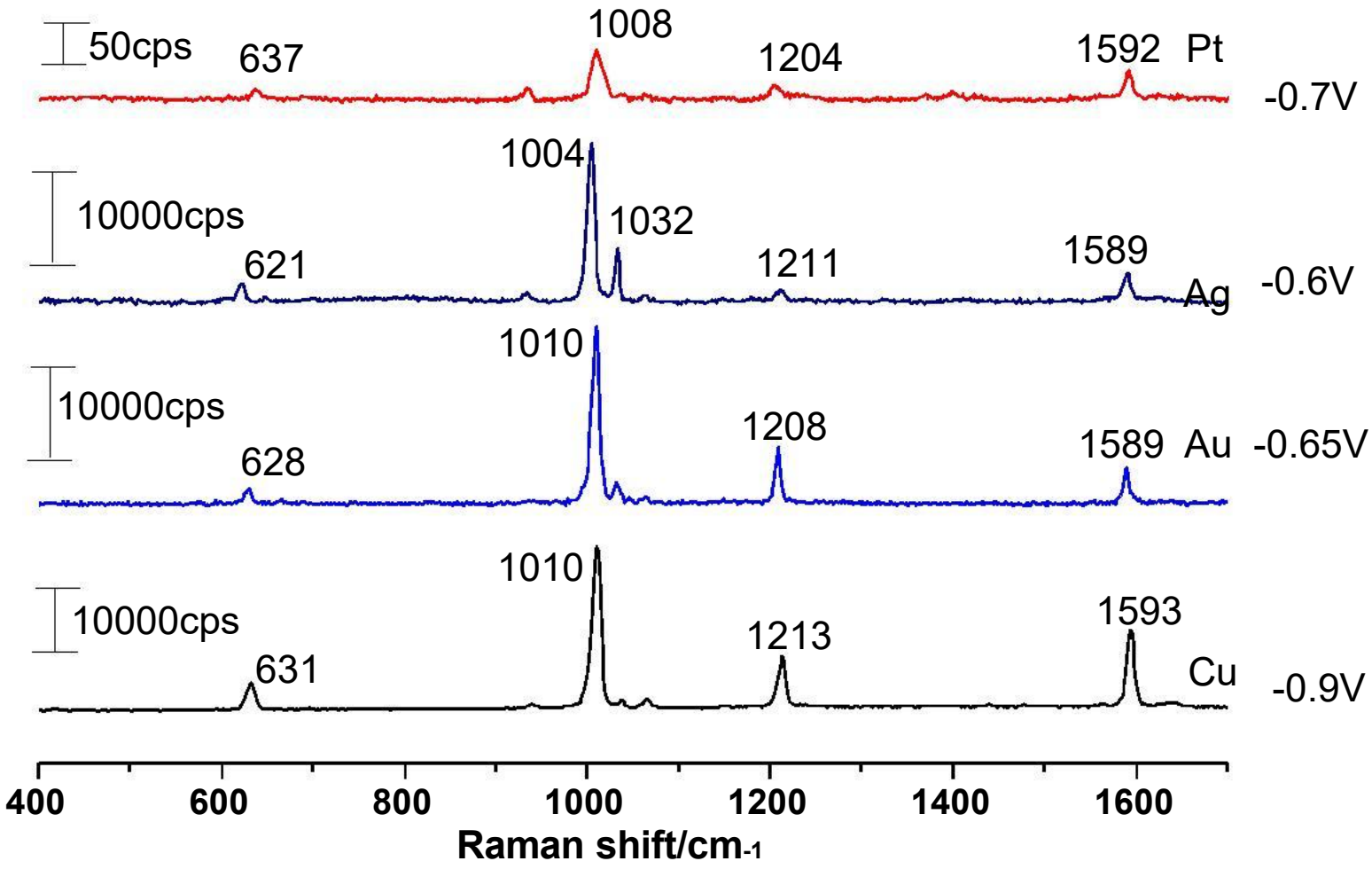
Sol. State Communi. 1982, 44,105

吡啶分子在不同金属电极上的**SERS**光谱



比较纯吡啶、吡啶水溶液（0.1M $NaClO_4$+0.1M Py ）以及吡啶吸附在粗糙的银、金、铜和铂表面（0.1M $NaClO_4$+ 0.01M Py）的拉曼光谱

峰值电位时吡啶SERS谱/Pt, Ag, Au and Cu (SCE)



SERS谱峰的频率和强度与金属和电极电位的关系？

电极/溶界面上的吸附自由能 ΔG_{ad}

1. 憎水项： 活性粒子自溶液体相移向界面层时以下二项的平衡

   - 减弱其对水分子短程有序结构的破坏程度而使体系自由能降低；

   - 活性粒子溶剂化程度减小而使体系自由能升高。

   带有极性基团有机化合物，如包含有碳氢链部分的（憎水部分）和易于水化的极性基团（亲水部分）的醇、醛、酸和胺等， 通常其憎水部分将朝向电极， 而亲水基团将朝向溶液一侧。

2. 电极表面与活性粒子之间的相互作用

   - 静电作用：金属表面与离子及偶极子、表面电荷与偶极子之间的相互作用；

   - 化学作用：活性粒子与电极表面发生部分电荷转移，吸附键的性质已接近化学键。

3. 吸附层中活性粒子之间的相互作用

   吸引力或排斥力，包括范德华力和静电作用，后者对带电离子相互作用更强；活性粒子覆盖度增加，作用力增加。

4. 置换电极表面上的水分子 - 水分子脱附能

双电层结构分析遵循的原则

1. 电极/溶液电势差由热力学决定, 状态确定后, 就有特定的数值; 而双电层的结构则与诸多因素有关.

2. 金属/电解液界面, 双电层结构主要由溶液一侧决定, 并可划分为紧密层和分散层. 紧密层厚度取决于离子的水化半径 (<1nm), 分散层厚度取决于溶液浓度 (几埃~几百纳米).

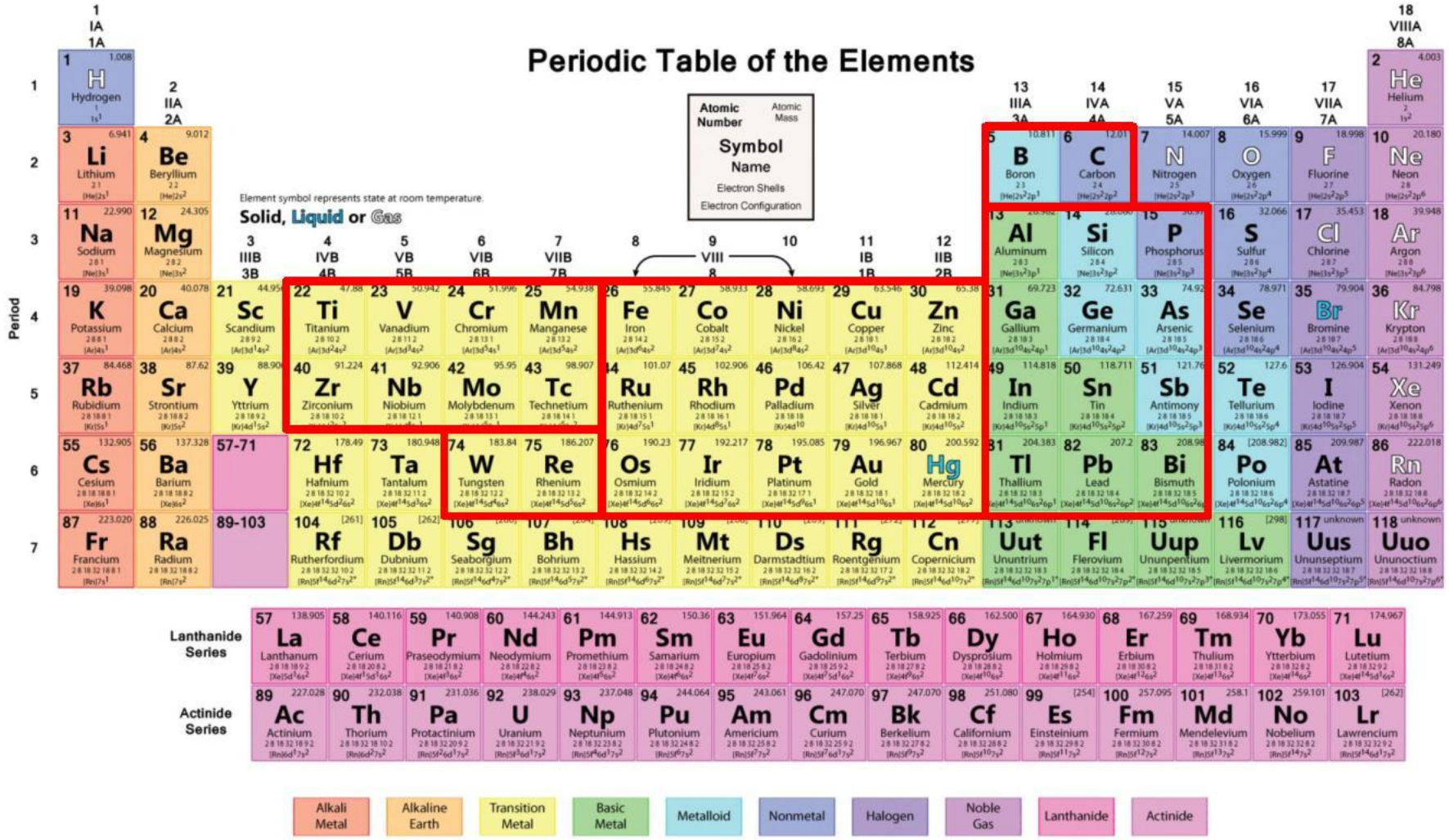
3. 分散层中的荷电粒子受长程静电相互作用和热运动制约, 并可用**Poisson-Boltzmann**方程建立界面电势分布的特征.

4. 紧密层因溶液中离子在电极上的吸附作用而产生的独特结构.

首先, 考虑离子的溶剂化层对紧密层厚度的影响, 以及极性溶剂分子介电饱和或偶极取向.

然后, 考虑离子的特性吸附造成紧密层中电势突变, 并可因发生阴离子超载吸附而形成三电层.

最后, 还要考虑有机分子吸附。


Periodic Table of the Elements
Atomic Number
Atomic Mass
Symbol
Name
Electron Shells
Electron Configuration
Element symbol represents state at room temperature.
Solid, Liquid or Gas
Period
1 IA 1A
2 IIA 2A
3 IIIB 3B
4 IVB 4B
5 VB 5B
6 VIB 6B
7 VIIB 7B
8 9 10 VIII 8
11 IB 1B
12 IIB 2B
13 IIIA 3A
14 IVA 4A
15 VA 5A
16 VIA 6A
17 VIIA 7A
18 VIIIA 8A
1 H Hydrogen
2 He Helium
3 Li Lithium
4 Be Beryllium
5 B Boron
6 C Carbon
7 N Nitrogen
8 O Oxygen
9 F Fluorine
10 Ne Neon
11 Na Sodium
12 Mg Magnesium
13 Al Aluminum
14 Si Silicon
15 P Phosphorus
16 S Sulfur
17 Cl Chlorine
18 Ar Argon
19 K Potassium
20 Ca Calcium
21 Sc Scandium
22 Ti Titanium
23 V Vanadium
24 Cr Chromium
25 Mn Manganese
26 Fe Iron
27 Co Cobalt
28 Ni Nickel
29 Cu Copper
30 Zn Zinc
31 Ga Gallium
32 Ge Germanium
33 As Arsenic
34 Se Selenium
35 Br Bromine
36 Kr Krypton
37 Rb Rubidium
38 Sr Strontium
39 Y Yttrium
40 Zr Zirconium
41 Nb Niobium
42 Mo Molybdenum
43 Tc Technetium
44 Ru Ruthenium
45 Rh Rhodium
46 Pd Palladium
47 Ag Silver
48 Cd Cadmium
49 In Indium
50 Sn Tin
51 Sb Antimony
52 Te Tellurium
53 I Iodine
54 Xe Xenon
55 Cs Cesium
56 Ba Barium
57-71
72 Hf Hafnium
73 Ta Tantalum
74 W Tungsten
75 Re Rhenium
76 Os Osmium
77 Ir Iridium
78 Pt Platinum
79 Au Gold
80 Hg Mercury
81 Tl Thallium
82 Pb Lead
83 Bi Bismuth
84 Po Polonium
85 At Astatine
86 Rn Radon
87 Fr Francium
88 Ra Radium
89-103
104 Rf Rutherfordium
105 Db Dubnium
106 Sg Seaborgium
107 Bh Bohrium
108 Hs Hassium
109 Mt Meitnerium
110 Ds Darmstadtium
111 Rg Roentgenium
112 Cn Copernicium
113 Uut Ununtrium
114 Fl Flerovium
115 Uup Ununpentium
116 Lv Livermorium
117 Uus Ununseptium
118 Uuo Ununoctium
Lanthanide Series
57 La Lanthanum
58 Ce Cerium
59 Pr Praseodymium
60 Nd Neodymium
61 Pm Promethium
62 Sm Samarium
63 Eu Europium
64 Gd Gadolinium
65 Tb Terbium
66 Dy Dysprosium
67 Ho Holmium
68 Er Erbium
69 Tm Thulium
70 Yb Ytterbium
71 Lu Lutetium
Actinide Series
89 Ac Actinium
90 Th Thorium
91 Pa Protactinium
92 U Uranium
93 Np Neptunium
94 Pu Plutonium
95 Am Americium
96 Cm Curium
97 Bk Berkelium
98 Cf Californium
99 Es Einsteinium
100 Fm Fermium
101 Md Mendelevium
102 No Nobelium
103 Lr Lawrencium
Alkali Metal
Alkaline Earth
Transition Metal
Basic Metal
Metalloid
Nonmetal
Halogen
Noble Gas
Lanthanide
Actinide

（6）新的模型及实验体系

（一）理论模型

超越平均场理论： 修正**PB**理论、**Ornstein-Zernike**积分理论、分子动力学、经典密度泛函理论…

静或动态第一性原理：量子化学和分子动力学

从头算、密度泛函、分子动力学、蒙特卡诺模拟

Monte-Carlo模拟

（二）新实验体系

界面：单、多层界面、多孔、自组装等新实验体系

电极材料：金属、半导体、液固体界面、生物

形貌、组成、大小(纳米)、形状、微观结构。。。

电解质：溶剂、液固体电解质、多组分。。。。

思考题 4（2026.10.8）

1. 写出**Frumkin**等温式，求出**g'=-2, 0, 2**时**σ-C** 曲线。相对于**Langmuir**等温式，并讨论吸引和排斥作用是如何影响等温曲线变化的。

2. 试通过**Poisson-Boltzmann**方程得到电化学界面在稀电解质（**1-1**型）溶液中分散层的计算电场值。

电场强度公式 $$\frac{d\phi}{dx} = -\left(\frac{8RTc_0}{\varepsilon_r\varepsilon_0}\right)^{1/2} sinh\left(\frac{zF\phi}{2RT}\right)$$

3.在无特性吸附时，试用**Gouy-Chapman-Stern**模型理解双电层电容**C**与紧密层C_H和分散层C_d电容之间的关系。

where ε is the dielectric constant of the medium, ε_0 is the permittivity of free space, and d is the interplate spacing.[2] The differential capacitance is therefore

$$\sigma = \frac{\varepsilon\varepsilon_0}{d}V \qquad \frac{\partial\sigma}{\partial V} = C_d = \frac{\varepsilon\varepsilon_0}{d} \tag{13.3.2}$$

[2]Here and elsewhere in this book we use the electrical relations appropriate to SI units, which lead to the following definition of Coulomb's law (24):

$$F = \frac{qq'}{4\pi\varepsilon\varepsilon_0 r^2}$$

The force F (in newtons) between two charges q and q' (in coulombs) is therefore related to the distance of charge separation r (in meters), the dielectric constant of the medium ε (dimensionless) and the *permittivity of free space* ε_0. The last parameter is a measured constant equal to $8.85419 \times 10^{-12}\,C^2N^{-1}m^{-2}$. This system has the advantage that the electrical variables are measured in common units. An alternative is the electrostatic system, where Coulomb's law is

$$F = \frac{qq'}{\varepsilon r^2}$$

The force F (in dynes) is related here to the charges (in statcoulombs) by the dielectric constant ε and the separation distance (in cm). Equations for the electrostatic system can be converted to corresponding relations for SI units by replacing ε with $4\pi\varepsilon\varepsilon_0$, and vice versa. Many treatments of interfacial structure involve electrostatic units. They are recognizable by the absence of ε_0 and the *appearance* of multiples of 4π in the results. In some treatments, $\varepsilon\varepsilon_0$ is denoted as a single quantity, usually ε, called the *permittivity* of the medium.